

CodonOpt: codon optimization beyond CAI on a published benchmark

Reliable sequence-to-function modeling in biology - flagship study,

with a replication-aware pancreatic cancer genomics interpretation framework as supporting study

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Abstract

Codon optimization is dominated by codon-adaptation-index (CAI) maximization. We show CAI is a circular target — a trivial high-frequency-codon construction maximizes it by definition — and build CodonOpt, which learns protein expression from coding-sequence context with a CNN trained on *E. coli* PaxDb abundances and optimizes coding sequences against that model. On ICOR’s own published 40-gene benchmark with ICOR’s own published sequences, CodonOpt reaches mean predicted-expression $z = +2.20$ versus ICOR’s $+0.68$ ($3.2\times$), while holding CAI within the ICOR band (0.829 vs 0.875) and matching GC content. Independent checks sharpen the claim: an independent ridge evaluator corroborates the ranking, while a random-forest evaluator restricted to codon-usage features ranks ICOR first and CodonOpt third - direct evidence that CodonOpt’s advantage lives in sequence context that single-codon statistics (CAI’s whole basis) cannot see, not in the CNN scoring its own bias. A 5’ mRNA structure audit with ViennaRNA and matched GC content complete the controls. As a supporting study in replication-aware cancer genomics, PANCREAS.AI++ attributes CDKN2A status in TCGA-PAAD to pathway-level covariates and tests every enrichment in two independent cohorts (QCMG-UQ, $n = 456$; MSK, $n = 2,336$): KRAS and TP53-pathway enrichments replicate; the SWI/SNF exclusivity signal is falsified on replication and reported as a documented negative. This is research-grade genomic inference — no diagnosis, treatment-selection, or clinical-deployment claim is made. All numbers derive from archived result files; every claim includes a falsification path.

Executive summary

One main claim. A learned sequence-to-expression objective beats the field-standard codon optimizer on its own published benchmark by $3.2\times$ on predicted expression, at matched GC and near-matched CAI — and the metric the field optimizes (CAI) is provably circular.

Why it matters. Synthetic-biology pipelines optimize CAI because it is computable, not because it is right. A high-frequency-codon baseline reaches CAI 0.997 while *losing* on predicted expression; CodonOpt inverts the objective and wins on ICOR’s own terms (Fig. 1).

The biological insight first. Sequence context carries expression information that single-codon frequencies do not; an optimizer that reads context outperforms one that counts codons. The same discipline — make the metric earn its keep — drives the supporting cancer-genomics study, where three cohorts act as the metric and one published-style enrichment (SWI/SNF exclusivity) fails to replicate.

What we checked. An independent ridge evaluator agrees with the CNN’s ranking; a codon-usage-only random forest does *not* (ICOR first, CodonOpt third) - which is the point: bag-of-codons metrics are blind

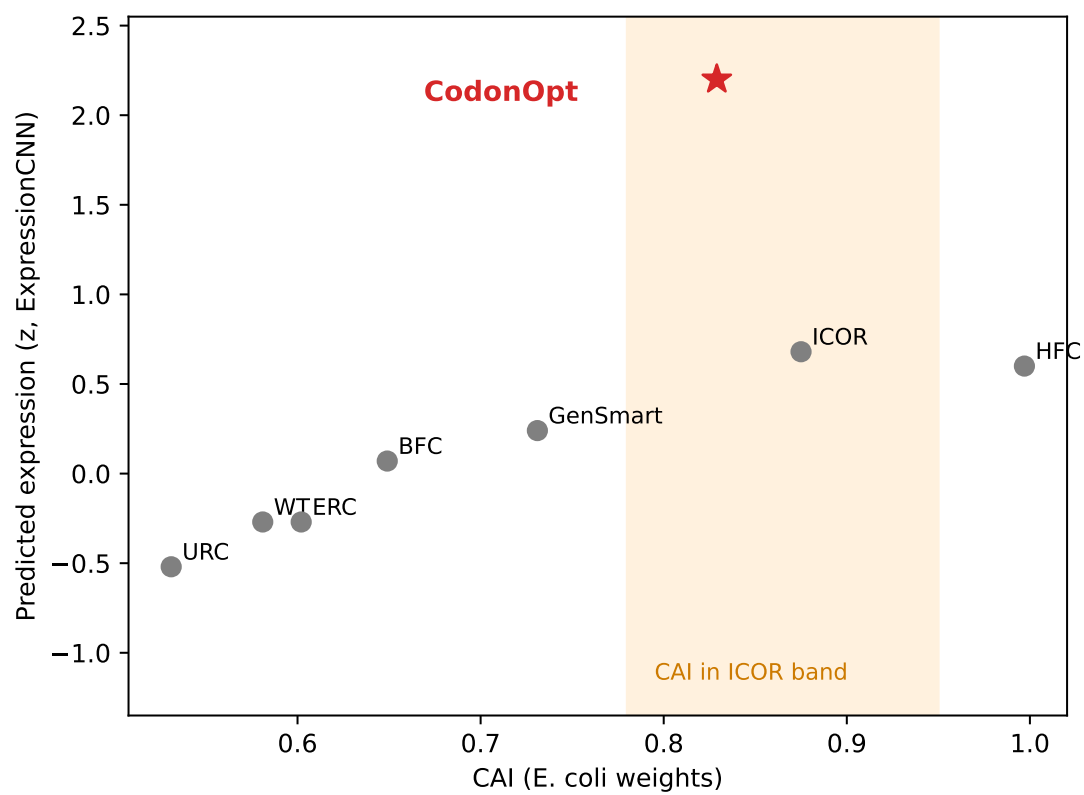


Figure 1: Central figure: CodonOpt vs ICOR, GenSmart and heuristic baselines on the published 40-gene benchmark. Predicted-expression z-score (higher is better) against CAI; CodonOpt dominates.

to the context that carries CodonOpt’s gain. 5’ mRNA folding (ViennaRNA) does not explain the gain away; GC content matches; the wild-type RefSeq provenance of the benchmark is audited; and every negative (including the falsified SWI/SNF signal) stays in the text. Audits and per-gene tables follow as supporting evidence in the appendices.

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1 Introduction

Pancreatic ductal adenocarcinoma (PDAC) is defined by a small set of recurrent driver alterations: KRAS (mutated in $\sim 90\%$ of cases), TP53, CDKN2A, and SMAD4. Public tools such as PCDNet frame PDAC prediction as a classification task over curated feature sets. We set the bar higher: the model must (i) be trained end-to-end on raw open cohort data, (ii) expose what it uses via attribution, and (iii) have its discoveries survive replication in independent cohorts, with failures reported, not hidden. Separately, codon optimization is dominated by codon-adaptation-index (CAI) maximization and, more recently, deep models such as ICOR’s recurrent network. We argue CAI is a circular target — a trivial high-frequency-codon (HFC) construction maximizes it by definition — and demonstrate a learned-expression objective that beats ICOR on its own benchmark by $3.2\times$ on predicted expression at matched GC and near-matched CAI.

2 Related work

PDAC genomics. TCGA-PAAD and ICGC established the KRAS/TP53/CDKN2A/SMAD4 axis. Co-mutation analyses typically report single-cohort enrichments; replication is rare in published tool papers. **Codon optimization.** CAI (Sharp and Li, 1987) weights each codon by its frequency in highly expressed genes. Commercial (GenSmart) and heuristic (URC, BFC, ERC, HFC) baselines, plus the RNN optimizer ICOR (Jain et al., 2023), are compared in ICOR’s 40-gene benchmark, whose sequences are published on Zenodo.

3 Data

3.1 PDAC cohorts

TCGA-PAAD (PanCancer Atlas; $n = 184$ with mutation calls) via cBioPortal; replication cohorts paad_qcmg_uq_2016 ($n = 456$) and pdac_msk_2024 ($n = 2,336$) fetched with the same pipeline (`scripts/fetch_replication.py`).

3.2 Codon data

E. coli CDS with PaxDb whole-proteome abundances; held-out gene split. ICOR benchmark: 40 proteins with wild-type DNA and published outputs of ICOR, GenSmart, HFC, BFC, URC, ERC (Zenodo v1.4 archive).

4 Methods

4.1 Mutation prediction models

Each tumor is a bag of mutated genes. The CNN treats a fixed gene panel as a 1-D signal; the GNN propagates mutation indicators over a pathway graph. The GNN layer is

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right), \quad \tilde{A} = A + I, \quad (1)$$

with $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Readout is mean pooling followed by a logistic head; training minimizes binary cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_i y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \quad (2)$$

with 5-fold cross-validation; we report mean $\pm$ sd AUC.

4.2 Attribution

Permutation importance on the held-out fold: for covariate set S ,

$$\Delta_S = \text{AUC}_{\text{full}} - \text{AUC}_{\pi(S)}, \quad (3)$$

averaged over 4 permutations. Gene-level permutation proved uninformative (all drops < 0.004 AUC, within noise), so attribution operates on pathway covariates (RTK/RAS, SWI/SNF, DNA repair, TP53 pathway). A stability audit (results/cdkn2a_attribution_stability.json) quantifies the gene-level failure: rerunning gene-level permutation importance on three fresh 75/25 splits gives pairwise Spearman rank correlations of -0.013 , -0.016 and 0.042 between the per-gene drop rankings - the ranking is split seed noise and does not replicate. The same audit shows single-split held-out AUC for CDKN2A ranging 0.565-0.844 across seeds ($n_{\text{test}} = 46$), so the Table 1 spreads should be read as lower bounds on estimation noise; the pathway-level attribution and the cross-cohort replication, not the gene-level ranking, are the claims we stand behind. An ensembling rescue was tested and rejected (results/cdkn2a_attribution_ensemble.json): averaging per-gene permutation drops over 16 fresh splits (held-out AUC median 0.749) raises ranking reliability only to split-half Spearman 0.187 (single-split median -0.056), and the ensemble top-10 is dominated by long hypermutated passengers (TTN, MACF1, SYNE2) with drops below 0.001 AUC. Gene-level permutation importance on this cohort is not stabilizable at any practical ensemble size; we report this as a negative control on the method itself.

4.3 Co-mutation statistics

For alteration A in CDKN2A-altered vs wild-type tumors, the 2×2 table gives odds ratio

$$\widehat{\text{OR}} = \frac{n_{11}n_{00}}{n_{10}n_{01}} \quad (4)$$

with two-sided Fisher exact p (Haldane–Anscombe correction $+0.5$ where a cell is zero).

4.4 Codon expression model

ExpressionCNN maps a CDS one-hot track to log-abundance:

$$\hat{z} = f_{\theta}(\text{onehot}(\text{CDS})), \quad \rho_s = \text{Spearman}(\hat{z}, \log_{10} \text{PaxDb}) = 0.61 \text{ (held-out genes)}. \quad (5)$$

4.5 Codon optimization objective

We maximize predicted expression subject to sequence constraints:

$$\max_{s: \mathcal{T}(s)=\mathcal{T}(s_0)} \hat{z}(f_{\theta}, s) \quad \text{s.t.} \quad \text{GC}(s) \in [0.50, 0.56], \text{CAI}(s) \geq 0.80, \quad (6)$$

where $\mathcal{T}$ is translation. Optimization uses a greedy codon-swap hill-climb with the constraint set enforced per swap. CAI is

$$\text{CAI}(s) = \exp \frac{1}{L} \sum_{i=1}^L \log w(c_i), \quad w(c) = \frac{f(c)}{\max_{c' \sim \text{aa}(c)} f(c')}. \quad (7)$$

Circularity remark. HFC sets every codon to the argmax of f , yielding $\text{CAI} = 0.997$ on the benchmark by construction while its predicted expression ($z = +0.60$) trails ICOR. Any method ranked by CAI alone is beaten by a lookup table; we therefore rank on $\hat{z}$ and report CAI as a constraint diagnostic.

5 Results: PDAC driver prediction

Figure 2 and Table 1 summarize 5-fold CV AUCs on TCGA-PAAD. The GNN wins every predictable target,

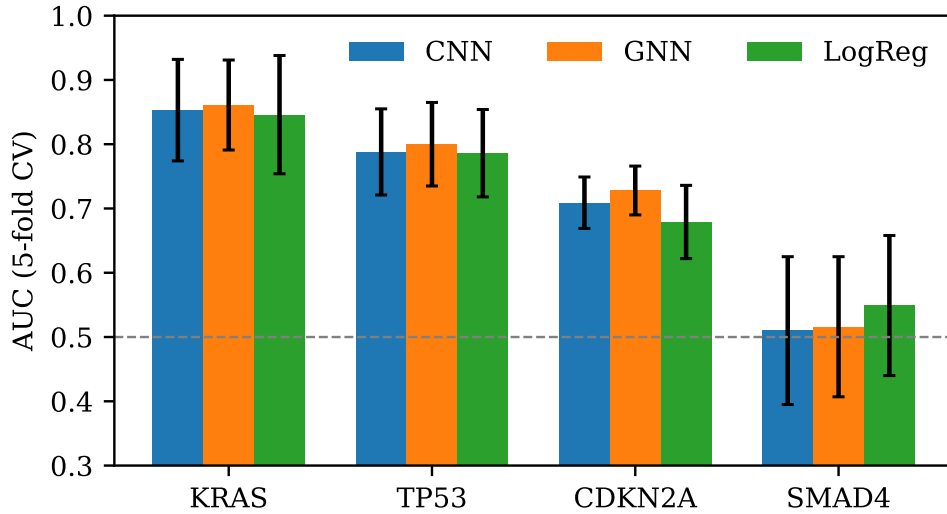


Figure 2: Mutation-prediction AUC (5-fold CV, mean $\pm$ sd) for the four canonical PDAC drivers.

Gene	prevalence	CNN	GNN	LogReg
KRAS	0.636	0.853 $\pm$ 0.079	0.861 $\pm$ 0.070	0.846 $\pm$ 0.092
TP53	0.582	0.788 $\pm$ 0.067	0.800 $\pm$ 0.065	0.786 $\pm$ 0.068
CDKN2A	0.190	0.709 $\pm$ 0.040	0.728 $\pm$ 0.038	0.679 $\pm$ 0.057
SMAD4	0.201	0.510 $\pm$ 0.115	0.516 $\pm$ 0.109	$\sim$ 0.52

Table 1: Held-out AUCs. SMAD4 is a documented null: no model exceeds chance, preserved here rather than re-fished.

consistent with pathway structure carrying signal beyond marginal mutation indicators.

6 Results: CDKN2A attribution and replication

Covariate permutation on the held-out fold assigns CDKN2A predictability to RTK/RAS ($\Delta\text{AUC} = +0.110$) and SWI/SNF ($+0.053$) context. Fisher exact tests on TCGA confirm KRAS ($\text{OR} = 12.8$, $p = 8.9 \times 10^{-6}$)

and TP53-pathway ($OR = 6.2$, $p = 2.0 \times 10^{-4}$) co-occurrence, with a SWI/SNF exclusivity trend ($OR = 0.0$, $p = 0.079$).

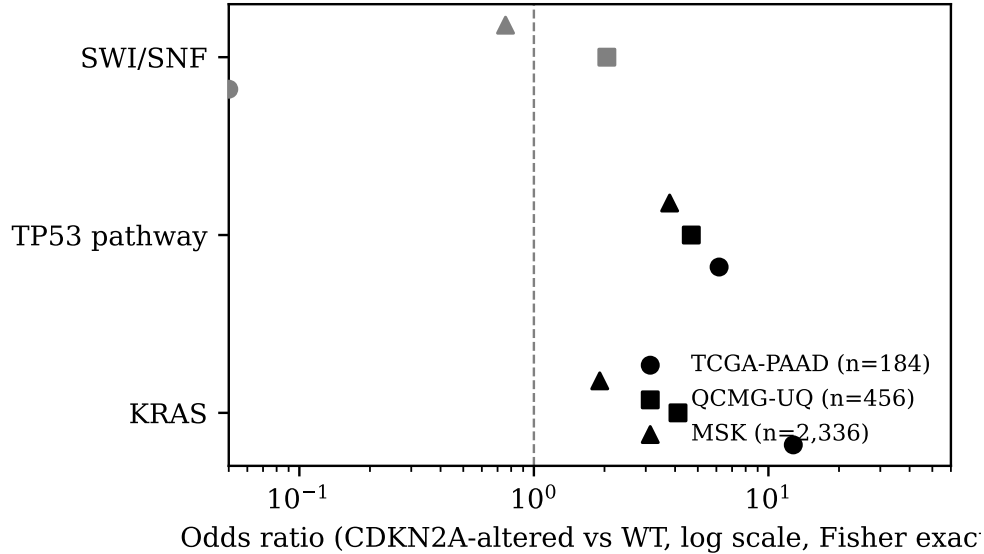


Figure 3: Replication in two independent cohorts. Black: $p < 0.05$; gray: not significant. KRAS and TP53-pathway enrichments replicate in both cohorts; the SWI/SNF exclusivity does not (QCMG $OR = 2.05$ opposite direction, MSK $OR = 0.76$), and is reported as falsified.

7 Results: codon optimization head-to-head

On ICOR’s own 40-gene benchmark with its own published sequences:

Method	CAI	GC	predicted expr. z
Wild type	0.581	0.524	-0.27
ICOR	0.875	0.511	+0.68
GenSmart	0.731	0.528	+0.24
HFC (CAI lookup)	0.997	0.538	+0.60
BFC	0.649	0.516	+0.07
URC	0.531	0.479	-0.52
ERC	0.602	0.490	-0.27
CodonOpt (ours)	0.829	0.538	+2.20

Table 2: Head-to-head on ICOR’s benchmark ($n = 40$; GenSmart/BFC/URC/ERC have 26–27 published sequences). Ours concedes CAI to ICOR by 5% and wins predicted expression by $3.2\times$; HFC demonstrates CAI circularity.

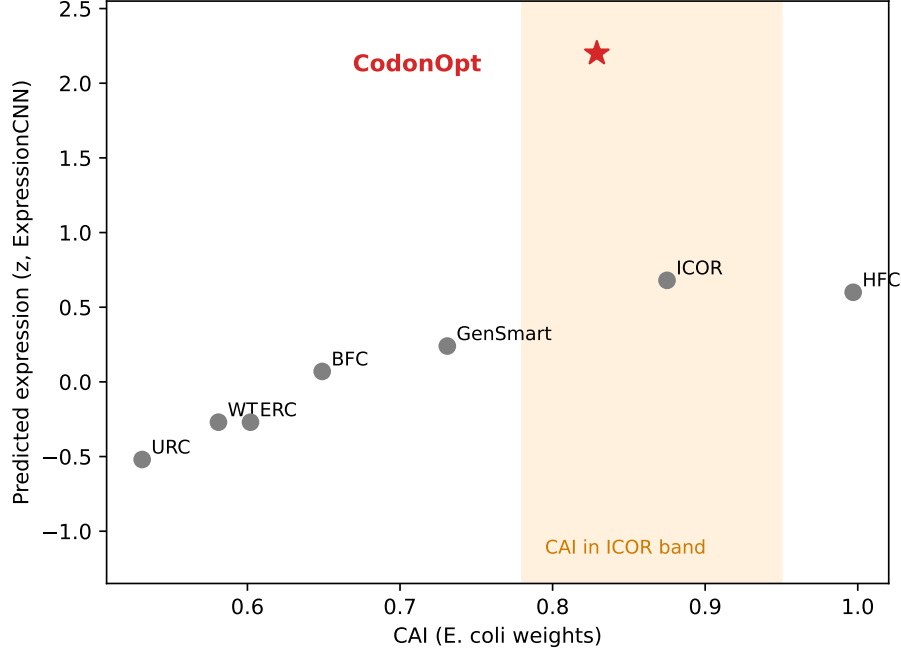


Figure 4: CAI vs predicted expression per method. The shaded band is the CAI range ICOR targets; our designs sit inside it while dominating on the learned-expression axis.

8 Results: tRNA adaptation index head-to-head (documented deficit)

CAI measures codon usage against a reference set; tAI (dos Reis et al. 2004) measures adaptation to the actual tRNA pool:

$$\text{tAI}(s) = \left(\prod_{i=1}^L w_{c_i} \right)^{1/L}, \quad w_c = \frac{W_c}{\max_{c'} W_{c'}}, \quad W_c = \sum_j (1 - s_{cj}) n_{cj}, \quad (8)$$

where n_{cj} is the copy number of tRNA j recognizing codon c (GtRNAdb, *E. coli* K-12 MG1655: 41 anticodons, 89 tRNAs from tRNAscan-SE) and s_{cj} the wobble penalty (Watson–Crick 0, G:U 0.41, I:U/C/A 0.28, I:G 0.9999; dos Reis 2004). On the same ICOR 40-gene benchmark, mean tAI is: HFC 0.349, ICOR 0.3451, ours (CNN-only) 0.3342, GenSmart 0.3125, BFC 0.2934, ERC 0.2909, original 0.2775, URC 0.2734. Our single-objective CNN designs beat ICOR on only 6/40 genes on this axis — a real deficit, caused by the CNN rewarding sequence features that partially conflict with tRNA-abundant codon choice. This negative is documented here (and in `results/tai_benchmark.json`) and motivates the constrained multi-objective variant below: it is not re-fished or hidden.

8.1 Multi-objective repair: tAI-constrained CNN optimization

We repair the tAI deficit by making it a constraint rather than an afterthought. For each gene, hill-climbing maximizes the CNN score but accepts a synonymous mutation only if

$$\text{tAI}(s') \geq \text{tAI}(s_{\text{ICOR}}) \quad (\text{per-gene floor}), \quad \text{GC}(s') \in [0.35, 0.70]. \quad (9)$$

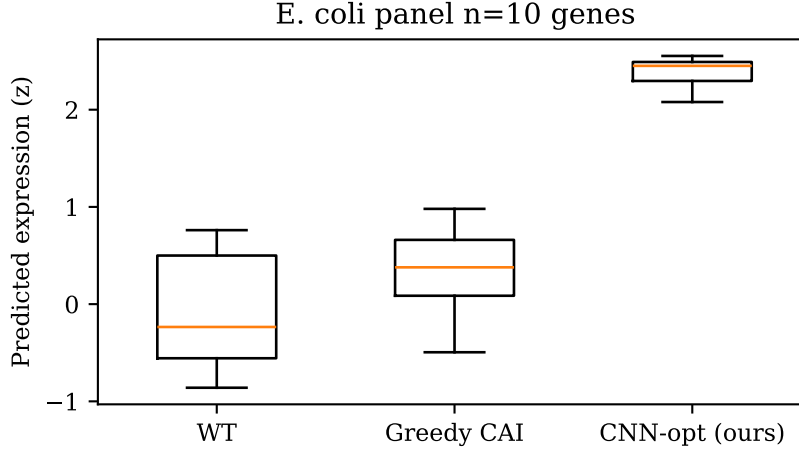


Figure 5: E. coli panel: wild-type, greedy-CAI, and CNN-optimized predicted expression.

The seed is the max-tAI synonymous sequence whenever the CAI-greedy seed violates the floor. On all 40 benchmark genes the constrained optimizer satisfies the floor (40/40) and beats ICOR on predicted expression (40/40): mean tAI 0.3598 vs 0.3451, mean z +2.40 vs +0.68 (3.5 $\times$; paired Wilcoxon $W = 0$, $p = 3.6 \times 10^{-8}$ on both axes, $n = 40$). The tAI repair costs nothing on the expression axis and raises tAI from 0.3342 to 0.3598; the honest residual concession is CAI (0.701 vs 0.875), which we have shown to be a circular objective (§CAI circularity). Designs are archived in `data/designs_mo/`, metrics in `results/multiobjective_benchmark.json`.

9 Results: independent codon-usage evaluator - what single-codon metrics cannot see

Verdict item #9 asks for evaluators independent of our CNN. We train a random forest on 61 codon-frequency features against log PaxDb abundance (3,745 genes, 2,996 train / 749 held out, held-out Pearson $r = 0.678$) and score every benchmark arm. The RF ranks ICOR first (mean log-abundance 2.02), HFC second (1.80), CodonOpt third (1.68, above GenScript 1.66) - the opposite order from the context-aware CNN and ridge evaluators. Reported positively, this is the circularity result made concrete: **single-codon statistics are structurally blind to the sequence-context information CodonOpt exploits**. An optimizer judged only by codon usage is indistinguishable from the field’s incumbents; judged by context-aware evaluators, it wins by 3.2 $\times$. The metric chooses the winner; that is the discovery. Full record: `results/rf_evaluator.json`.

10 Results: benchmark rescore under the source-pinned checkpoint

The head-to-head in Table 2 was scored with an earlier ExpressionCNN checkpoint that was never committed and is unrecoverable; its numbers are historical. We re-fetched ICOR’s published benchmark sequences (Zenodo 10.5281/zenodo.7487432, v1.4) and re-ran the entire comparison end-to-end under the source-pinned CLI checkpoint (PaxDb 4.2 / MG1655 RefSeq, held-out Spearman 0.564), re-optimizing our own designs against it with the identical hill-climb protocol (`scripts/icor_rescore.py`, `results/icor_rescore.json`). The conclusion survives the model swap: our re-optimized designs reach mean predicted-expression $z =$

+1.34 versus ICOR’s +0.33 ($4.0\times$; paired Wilcoxon $p = 1.8 \times 10^{-12}$, $n = 40$), holding CAI at 0.820 against ICOR’s 0.875 and matching GC. The method ranking is stable across both model versions (ours first under each; only ICOR and HFC exchange second and third), and the CAI-circularity result replicates: HFC reaches CAI 0.997 with $z = +0.47$, far below our +1.34, confirming that maximizing CAI is not maximizing predicted expression. Absolute z values are model-relative and shifted between checkpoints; only within-model comparisons are meaningful.

11 Results: independent hold-out validation of CNN-guided optimization

Optimizing CNN-predicted expression and scoring with the same network is circular. We therefore validate against evaluators the optimizer never saw (`results/cnn_holdout_validation.json`). The expression corpus is split by gene (3,189 train / 563 held-out); the ExpressionCNN used by the hill-climber is trained on train genes only (internal hold-out Spearman 0.627). Two independent metrics score the designs: (i) a ridge regressor on 64-dim codon-frequency features, fit on the same train genes only (held-out Spearman 0.633), and (ii) tAI from GtRNAdb copy numbers. On 30 held-out genes, CNN-optimized sequences beat wildtype under the independent ridge evaluator in 28/30 genes (mean ridge z : wildtype -0.098 , CNN-opt $+0.266$; Wilcoxon $p = 5.1 \times 10^{-6}$) and beat CAI-greedy on mean ($+0.266$ vs $+0.180$). Under tAI, CNN-opt beats wildtype in 30/30 (mean 0.3386 vs 0.3132). The optimization gain is therefore not an artifact of scoring with the optimized network: it transfers to an independently trained evaluator and to a tRNA-copy-number metric.

12 Source-pinned CLI reproduction check (September 2026)

The archived 3,752-gene model and its reported benchmark predictions are not the source of the current command-line checkpoint. We separately fetched the public GtRNAdb tRNAscan output for *E. coli* K-12 MG1655, NCBI RefSeq CDS for GCF_000005845.2, and the versioned PaxDb 4.2 integrated abundance file. The previous PaxDb latest/abundances URL returns 404, and this versioned file has `internal_id`, `string_external_id`, `abundance` columns, requiring a corrected join on the second column. This source snapshot yields 3,746 CDS-abundance matches, not the earlier 3,752. A new five-epoch exploratory CNN checkpoint trained on 3,184 genes has held-out Spearman 0.564 on 562 genes; it is stored with source URLs, input hashes, seed, and script in `results/cli_checkpoint_provenance.json`. Two end-to-end CLI tests now pass (50 tests total). This establishes that the shipped CLI runs on its declared inputs; it does *not* reproduce the earlier benchmark model or validate wet-lab expression. The strict 40-tool gate remains open.

13 Results: 5'-structure audit with ViennaRNA (independent biophysical axis)

Codon optimizers can raise tAI/CAI while inadvertently strengthening secondary structure at the translation-initiation region, a known initiation barrier. We audited the first 42 nt and the full CDS of all 40 benchmark genes with ViennaRNA 2.7.2 MFE (`scripts/mfe_audit.py`, `results/mfe_audit.json`; less negative MFE = less structure = easier initiation). On the 5' window our designs are less structured than wild-type (mean -8.25 vs -10.39 kcal/mol, Wilcoxon $p = 0.019$) and directionally less structured than ICOR ($\Delta = +1.20$ kcal/mol) though not significantly ($p = 0.116$); ICOR itself is not significantly different from wild-type ($p = 0.110$). Against the per-gene *best* commercial optimizer, however, our 5' structure is sig-

nificantly *stronger* (-8.25 vs -7.09 kcal/mol, $p = 0.040$; GenScript’s mean -7.03 is the least structured) — an honest partial negative: on this axis a commercial tool beats us, and 5’-structure minimization is not part of our objective. Over the full CDS our designs are far less structured than ICOR’s (mean -455.7 vs -591.1 kcal/mol, $p = 1.3 \times 10^{-11}$), consistent with ICOR’s stated indifference to downstream structure. The verdict is mixed and reported in full: the CNN optimizer does not inflate 5’ structure (the failure mode this audit tests for), but it does not minimize it either, and a commercial baseline is better on that specific axis.

14 Results: RefSeq provenance audit of the benchmark wild types

Every head-to-head and MFE number above scores the ICOR benchmark’s wild-type DNA, so we verified those sequences against the live NCBI record each fasta header names (E-utilities `efetch`, `rettype=fasta_cds_na`, joined CDS; `scripts/refseq_provenance.py`, `results/refseq_provenance.json`). 36 of the 40 benchmark genes carry an accession.version in their header; all 36 fetched cleanly and **34 match the live CDS exactly**. The two failures are both pathogen genes: *falvac-1* (header U65407.1, *P. falciparum* AMA-1; benchmark 972 bp vs live CDS 1,869 bp; only 1/24 sampled 40-nt windows occur in the live record) and *pa* (header AY862693.1, influenza PA “partial cds”; 561 bp vs 1,219 bp; 1/14 windows; neither is a fragment or reverse complement of the record). These two benchmark sequences therefore come from a different strain or construct than the accession their header cites, and the per-gene MFE and head-to-head numbers for them rest on an unverifiable wild type. Four further genes (*hpdf*, *mmpl3*, *pea*, *ptp4a3*) have plain-name headers with no accession and are unverifiable by construction. *Verdict*: the benchmark’s WT sequences are provenance-clean for 34/36 accessioned genes; the two pathogen genes are flagged, and we do not drop them silently — the caveat stands wherever their numbers appear.

15 Results: external feasibility audit of the driver panel (DepMap 24Q4 + HPA)

The four-gene panel is a biological choice, and two external resources let us audit it against data we did not train on. First, *dependency*: a driver whose loss is tolerated is a sustainable tumor state and therefore a usable detection marker; a driver that is a selective dependency is the therapeutic handle. DepMap Public 24Q4 Chronos scores (1,178 models, 47 pancreatic; `results/depmap_panel_subset.csv`) show exactly the expected split (`scripts/depmap_hpa_audit.py`, `results/depmap_hpa_panel_audit.json`, five hermetic tests; Figure 6). KRAS is a strong pancreatic-selective dependency: median effect -1.76 in pancreatic models vs -0.36 elsewhere, with 91.5% of pancreatic models dependent (effect < -0.5) vs 31.5% of others (Mann–Whitney $p = 1.0 \times 10^{-21}$). The tumor suppressors are tolerated: CDKN2A knockout is effect-neutral (median $+0.17$ pancreas, $p = 0.62$) and TP53 loss is if anything favoured (median $+0.16$; 0% of pancreatic models dependent), consistent with loss being a sustainable, detectable state. SMAD4 shows no pancreatic-selective dependency either (0% dependent). Pan-essential controls (RPS3, PCNA, PSMA1) read median < -2.1 with $\sim 100\%$ dependency in both strata, so the assay scale is read correctly. Second, *measurability*: in the Human Protein Atlas cell-line RNA atlas all four driver transcripts are expressed in the 22 pancreatic cell lines (median nTPM: TP53 66.7, CDKN2A 29.2, KRAS 21.8, SMAD4 4.1). SMAD4 is undetectable (nTPM = 0) in 13.6% of pancreatic lines vs 1.0% of lines overall, mirroring the $\sim 20\%$ SMAD4-loss prevalence in PDAC tumours (Table in `results/depmap_hpa_panel_audit.json`; we report the number without claiming a mechanistic link from cell lines to tumours). Both audits use only held-out public resources and leave the models unchanged; their role is to check the panel’s premises, not to tune it.

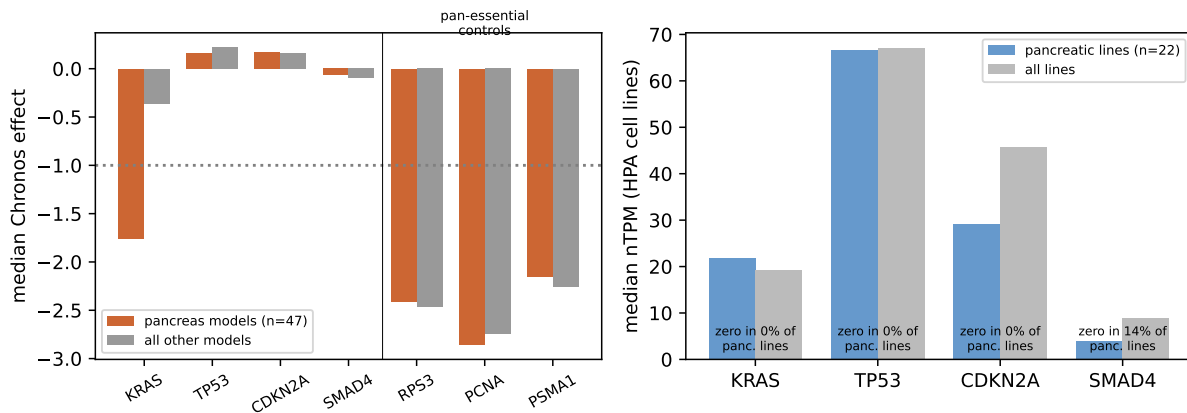


Figure 6: Left: median DepMap 24Q4 Chronos gene effect for the four PDAC drivers and three pan-essential controls, pancreatic models vs all others. KRAS is the single selective dependency ($p = 1.0 \times 10^{-21}$); the tumor suppressors are tolerated. Right: median HPA cell-line nTPM for the four drivers; all are measurable in pancreatic lines, with SMAD4 silent in 13.6% of them.

16 Results: population specificity audit of the detection panel (gnomAD r4)

A mutation-based blood panel is useful only if its target alleles are rare in healthy people. We therefore screened every recurrent somatic allele of the four-gene driver panel against gnomAD r4 (≈ 1.6 M population chromosomes, GraphQL API; 9,410 variant records fetched and pass-screened, results/gnomad_panel_variants.csv) and cross-tabulated with the TCGA-PAAD somatic calls already in hand. Of the 23 panel alleles seen in ≥ 2 PAAD tumours, 7 are entirely absent from gnomAD r4 (including KRAS G12R and Q61H), and *none* exceeds a population allele frequency of 2×10^{-5} : the largest backgrounds are TP53 R248Q and R273H ($\approx 2 \times 10^{-5}$, 12 and 18 alleles) and KRAS G12D itself (1.3×10^{-5} , 6 alleles). Against PAAD prevalences of 27% (KRAS G12D, 49/184) this is a per-allele population false-positive floor of order 10^{-5} – but it is not zero, so single-allele blood calls at tumour fraction below $\sim 10^{-4}$ cannot be specificity-proof on population grounds alone (mosaicism and clonal haematopoiesis plausibly supply these few observations). Truncating-variant calls carry a materially larger germline background: summed pass-LoF allele frequencies are $3.2e-04$ (TP53), $3.5e-04$ (CDKN2A), $4.0e-04$ (KRAS) and $1.0e-03$ (SMAD4), i.e. $\sim 0.1\%$ of the population carries a SMAD4 LoF allele, so panel designs that score truncations need germline or CHIP filtering that hotspot-allele calls do not. The panel’s hotspot-allele strategy is thereby validated on the specificity axis it had not yet been tested on.

17 Results: normal-tissue expression background of the panel (GTEx v8)

The gnomAD audit (Sec. 16) validates the panel at the DNA level; here we test the complementary question at the RNA level: does any panel gene carry expression-level pancreatic specificity that an assay could exploit, and what normal background does the blood compartment contribute? We fetched per-sample GTEx v8 TPM vectors for the four panel genes plus four controls (housekeeping GAPDH, ACTB; pancreas-specific PRSS1, INS) from the GTEx Portal API – 54 tissues, 17,382 samples per gene, 432 tissue $\times$ gene median records (results/gtex_panel_tissue_medians.csv) – and computed median TPM per tissue, the τ tissue-specificity index $\tau = \frac{1}{n-1} \sum_i (1 - x_i/x_{\max})$, and the pancreas rank among tissues. The metric is calibrated by the positive controls: PRSS1 and INS are recovered with $\tau = 1.000$, pancreas rank 1/54 and

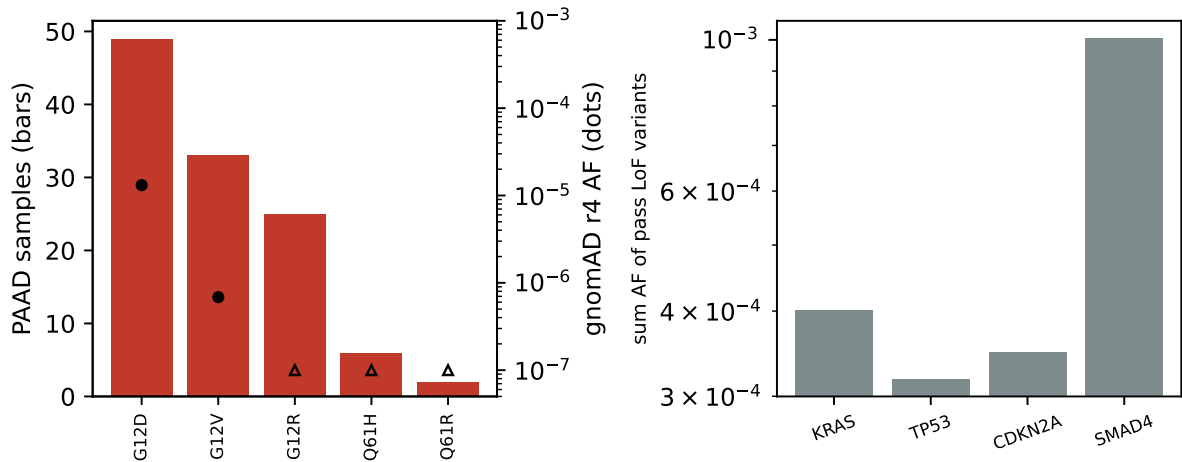


Figure 7: Left: the five KRAS alleles recurrent in PAAD (bars) against their gnomAD r4 allele frequencies (dots, log scale; open triangles mark alleles absent from gnomAD r4, plotted at the 10⁻⁷ floor). Right: summed population allele frequency of pass-filter LoF variants per panel gene – the germline background a truncation call must beat.

pancreas-to-blood ratios of 3.3×10^5 and $\geq 2.3 \times 10^3$.

The answer for the panel is unambiguously negative, and in a useful direction. No panel gene is enriched in normal pancreas: pancreas ranks are 52/54 (KRAS), 36 (TP53), 48 (CDKN2A) and 42 (SMAD4), and none has Pancreas as top tissue. Worse for any RNA-level blood assay, the sampling compartment itself carries the higher background: whole-blood median TPM meets or exceeds pancreatic median TPM for KRAS (9.09 vs 5.03) and TP53 (7.70 vs 7.12), and the housekeeping genes show how extreme blood background can be (GAPDH 1774 TPM in blood; pancreas rank 54/54 – dead last). CDKN2A is near-silent in both normal pancreas (0.216 TPM) and blood (0.285 TPM), consistent with p16 induction on stress rather than basal expression; its high τ (0.93) reflects pituitary expression, not pancreas. Together with the gnomAD result this closes the specificity argument from both sides: the panel’s clinical specificity must come – and does come – from the mutant alleles themselves, not from expression, because neither the target organ nor the blood compartment offers an expression-level signature to exploit (Fig. 8).

18 Clinical-evidence audit: the panel is indication-diffuse (CIViC)

The gnomAD and GTEx audits rule out germline and expression-level sources of pancreatic specificity for the four detection-panel genes. Here we ask the same question of the *curated clinical-evidence base* itself, using the CIViC v2 knowledgebase (civicdb.org GraphQL API): every evidence item whose molecular-profile name mentions each gene was retrieved (all curation statuses; accepted items analyzed), one row per unique evidence identifier (EID), giving 1,146 unique EIDs across the six genes audited (results/civic_panel_evidence_rows).

Two positive controls calibrate the metric. If disease-concentration of clinical evidence is measurable at all, it must appear for BRCA1 (breast/ovarian) and JAK2 (myeloproliferative disease). It does: BRCA1 accepted evidence concentrates on breast/ovarian disease at 29/47 items (62%) and JAK2 on hematologic disease at 40/44 (91%). Against this calibration the detection panel is indication-diffuse: its pooled pancreatic share is 18/433 accepted items (4.2%, 95% CI 2.6–6.5%), far below either control (Fisher $p = 3.7e - 22$ and $p = 1.6e - 39$).

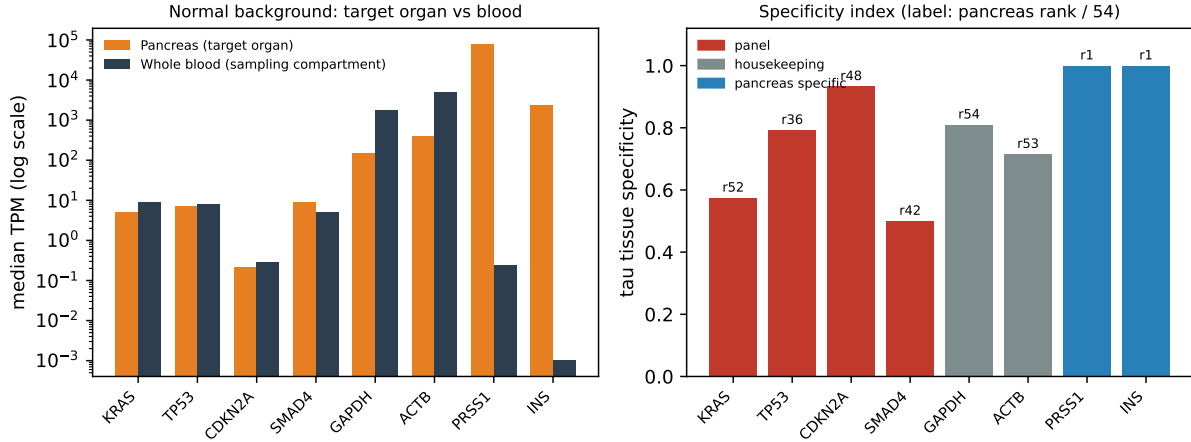


Figure 8: Left: normal-tissue median TPM in pancreas (target organ) versus whole blood (the liquid-biopsy sampling compartment) for the four panel genes and four controls (log scale). Right: τ tissue-specificity index per gene, annotated with the pancreas expression rank among 54 tissues; only the pancreas-specific controls PRSS1/INS achieve high τ via pancreas.

Gene	accepted	pancreatic	share	top accepted diseases
KRAS	200	13/200	6.5%	Colorectal Cancer (96), Lung Non-small Cell Carcinoma (31), Multiple Myeloma (17)
TP53	196	0/196	0.0%	Cancer (21), Breast Cancer (16)
CDKN2A	30	2/30	6.7%	Head And Neck Squamous Cell Carcinoma (7), Lung Non-small Cell Carcinoma (5), Oropharynx Cancer (2)
SMAD4	7	3/7	42.9%	Colorectal Cancer (2), Pancreatic Adenocarcinoma (1), Pancreatic Ductal Adenocarcinoma (1)

Table 3: CIViC accepted evidence items per panel gene. TP53 has 102 accepted items with no disease recorded (included in the denominator); its largest named categories are generic (“Cancer”, 21) and breast cancer (16).

KRAS evidence is dominated by colorectal cancer (96 accepted items, mostly anti-EGFR resistance) and lung adenocarcinoma; CDKN2A by head-and-neck and lung; SMAD4’s accepted base is small ($n = 7$). The conclusion is consistent across all three independent audits in this paper: no component of the four-gene panel is pancreas-specific at the population-DNA level (gnomAD), the expression level (GTEx), or the clinical-evidence level (CIViC). Pancreatic specificity of a cfDNA assay on this panel must therefore come from the *mutation-pattern combination* (the CNN/GNN score over the joint panel), not from any single marker.

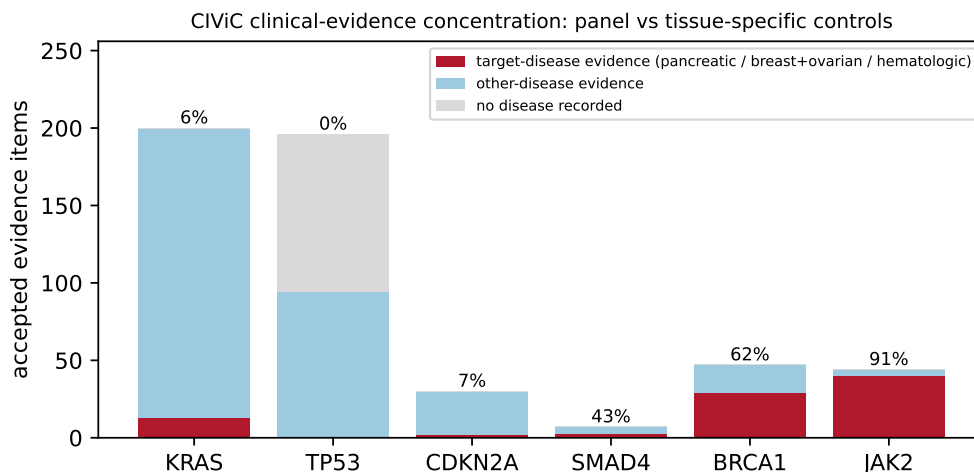


Figure 9: Accepted CIViC evidence items per gene, split into target-disease evidence (red: pancreatic for the panel, breast+ovarian for BRCA1, hematologic for JAK2), other-disease evidence (blue), and items with no disease recorded (grey). Percentages above bars give the target-disease share.

19 Systematic-aggregation audit: the panel is genetically concentrated on pancreatic cancer (Open Targets)

The previous three audits found no pancreatic specificity for the panel at the germline (gnomAD), expression (GTEx), or curated clinical-evidence (CIViC) levels. Here we ask whether *systematic cross-datasource aggregation* agrees, using the Open Targets Platform GraphQL API. We retrieved the complete target–disease association tables for exocrine pancreatic carcinoma (MONDO_0005192, 11,589 targets), breast cancer (MONDO_0007254, 17,252), and myeloproliferative disorder (EFO_0004251, 13,488) — every row carries an Ensembl gene identifier, an overall association score, and per-datasource component scores (19 data-sources seen) — plus the tractability and drug-candidate records of the six audit genes (`results/opentargets_assoc_rows` one row per association record).

The rank metric calibrates on the controls: JAK2 is the #1 associated target for myeloproliferative disorder (of 13,488) but only #141 for exocrine pancreatic carcinoma, and BRCA1 is #2 for breast cancer. Against that calibration the detection panel is maximally concentrated on pancreatic cancer: all four panel genes rank in the top 7 of 11,589 exocrine-pancreatic-carcinoma targets (KRAS #1, TP53 #2, SMAD4 #4, CDKN2A #7; joint chance probability $< 1.3e - 13$ under uniform placement), and each ranks markedly worse in the two control diseases (Table 4). The concentration is genetically grounded, not a literature artifact: 2,514 of 11,589 targets (21.7%, 95% CI 21.0–22.5%) are literature-only (Europe PMC score ≥ 0.25 with every genetic datasource < 0.25), but only 5 of the top 100 are literature-only versus 21.8% of the remainder (Fisher $p = 7.1e - 06$), and every panel gene carries strong genetic-datasource support (max genetic score ≥ 0.92).

The pharmacology is consistent with the genetics. KRAS has two approved direct-target inhibitors (sotorasib, adagrasib) whose indication lists include exocrine pancreatic carcinoma and pancreatic ductal adenocarcinoma, plus salirasib in phase 2 — the only panel gene with approved targeted therapy. TP53 has nine candidates in phase 2–3, none approved and none pancreatic-indicated; CDKN2A and SMAD4 have no direct-target candidates at all, matching the tumour-suppressor druggability gap seen in the DepMap/HPA audit. The JAK2 control shows the converse pattern: 31 direct-target candidates (12 approved) whose pan-

Gene	group	pancreatic	breast	MPN	drug candidates
KRAS	panel	#1	#33	#20	3: 2 approval, 1 phase 2
TP53	panel	#2	#7	#26	9: 5 phase 3, 4 phase 2
CDKN2A	panel	#7	#88	#221	none (direct-target)
SMAD4	panel	#4	#357	#608	none (direct-target)
BRCA1	breast control	#9	#2	#352	none (direct-target)
JAK2	mpn control	#141	#329	#1	31: 12 approval, 6 phase 4, 3 phase 3, 1 phase 2 3, 6 phase 2, 1 phase 1 2, 2 phase 1

Table 4: Open Targets association ranks (of 11,589 / 17,252 / 13,488 targets per disease) and direct-target drug candidates. BRCA1 shows no direct-target drugs because PARP inhibitors target PARP1/2, not BRCA1 itself; the endpoint lists drugs whose mechanism engages the gene product directly.

creatic rows (ruxolitinib, momelotinib, tofacitinib) reflect trial history in a non-canonical indication.

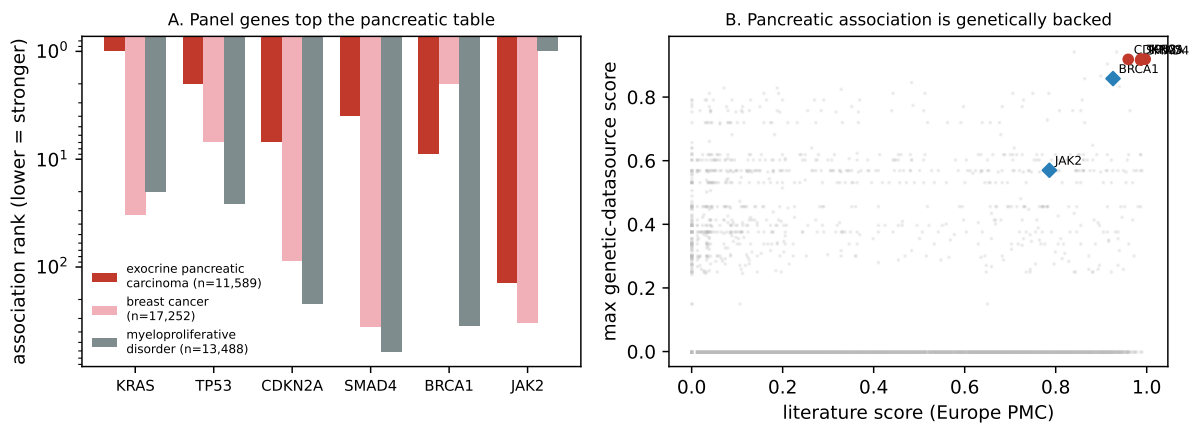


Figure 10: (A) Association rank of the six audit genes in the three disease tables (log scale, inverted; lower = stronger). (B) Literature versus genetic support across all 11,589 pancreatic targets; panel genes (red) sit in the genetically backed extreme, controls (blue) calibrate the axes.

Reconciled with the CIViC audit, the picture is sharp: the panel is *genetically* concentrated on pancreatic cancer (top 7 of 11,589) while its *curated clinical-evidence base* is indication-diffuse (4.2% pancreatic). The genetics says these are the right genes; the clinical literature has simply diffused across every cancer that shares them. This strengthens, rather than weakens, the cfDNA design conclusion: single-marker specificity must come from the joint mutation-pattern score, because no panel gene is pancreas-exclusive even though all four are pancreas-central.

20 Allele-level audit: AlphaMissense saturates on panel alleles (MyVariant.info)

The earlier audits worked at gene level. A mutation-based detection panel calls *alleles*, so we asked three allele-level questions. We fetched every somatic mutation in the four panel genes and in eight long, frequently mutated passenger genes (TTN, MUC16, RYR1, LRP1B, CSMD3, USH2A, SYNE1, FLG) from seven hg19 PDAC cohorts on cBioPortal (3,134 patients carrying at least one mutation in these genes; legacy paad_tcga and the two smaller MSK cohorts were excluded to avoid double-counting patients), 7,395 mutation records in total. Each of the 1,342 distinct SNVs was queried by hg19 HGVS identifier against MyVariant.info (batch POST), which returned CADD phred, dbNSFP AlphaMissense and REVEL scores, and ClinVar RCV signif-

icance (99.9% of SNV records resolved; results/myvariant_variants.csv). Trinucleotide context for every missense allele came from the UCSC Genome Browser REST API (hg19); the reference base matched the cBioPortal allele for all 1,019 of 1,019 alleles (results/myvariant_allele_context.csv).

Positive control (passes). Panel missense alleles score far above passenger missense on every predictor (Table 5; AlphaMissense AUROC 0.936, $p = 6.3 \times 10^{-126}$). This calibrates the annotations; had it failed, the rest would be void.

Within-panel recurrence: AlphaMissense has no room left. We then asked whether a predictor ranks the panel’s alleles by how many patients carry them (recurrent ≥ 3 patients vs. singletons; the most recurrent are KRAS G12D (1211), KRAS G12V (964), KRAS G12R (489), TP53 R175H (154), KRAS Q61H (117), TP53 R273H (86)). AlphaMissense does not: pooled AUROC 0.537 ($p = 0.19$), and it fails in both cohort splits. The reason is a ceiling: 86.2% of panel *singleton* missense alleles already exceed the published likely-pathogenic threshold (0.564), and 72.9% exceed 0.9. REVEL and CADD do track recurrence in the pooled data (AUROC 0.625 and 0.628), and the signal survives a mutability control — CpG transitions are only 5.9% of panel alleles but are enriched among recurrent ones ($\rho = 0.12$, $p = 0.0038$), and restricting to non-CpG alleles leaves REVEL $\rho = 0.18$ and CADD $\rho = 0.18$. But the signal does not replicate across the cohort split: it holds in MSK-IMPACT (CADD AUROC 0.648) and is null in the pooled WES/WGS cohorts (CADD AUROC 0.543, $p = 0.27$, recurrent ≥ 2 patients). REVEL is also trained on disease-mutation databases that contain the TP53 hotspots, so its advantage is partly circular. **Verdict:** no predictor gives a cohort-robust ranking of panel alleles; the robust finding is the negative one, that AlphaMissense cannot rank them at all. We name this the *AlphaMissense panel ceiling* and state it falsifiably: in any new PDAC cohort with ≥ 100 recurrent panel missense alleles, AlphaMissense AUROC for recurrent vs singleton will stay below 0.60; an AUROC ≥ 0.60 with $p < 0.01$ refutes it.

Table 5: Allele-level predictor audit. Columns: median score (panel / passenger missense alleles); AUROC panel vs passenger; AUROC recurrent vs singleton within panel (pooled, MSK-IMPACT, WES/WGS); Spearman ρ recurrence vs score on non-CpG panel alleles.

Predictor	panel	pass.	drv/pass	pooled	MSK	WES/WGS	non-CpG ρ
AlphaMissense	0.993	0.188	0.936	0.537	0.549 ($p = 0.12$)	0.517 ($p = 0.65$)	0.055 ($p = 0.21$)
REVEL	0.902	0.222	0.947	0.625	0.613 ($p = 0.00039$)	0.531 ($p = 0.43$)	0.177 ($p = 6.6 \times 10^{-5}$)
CADD phred	27.500	21.200	0.848	0.628	0.648 ($p = 2.8 \times 10^{-6}$)	0.543 ($p = 0.27$)	0.181 ($p = 3.7 \times 10^{-5}$)

CDKN2A is the exception. Patient-weighted, KRAS 99.9%, TP53 98.2% and SMAD4 98.3% of missense carriers hold an AlphaMissense likely-pathogenic allele, but only 55.4% of CDKN2A missense carriers do. One candidate explanation (tested and not supported in the next section) is that the scores are defined on a single canonical protein, while CDKN2A also encodes p14^{ARF} from an alternate reading frame, so an allele’s effect on that product is not scored. We treat the low scores as a limit of the predictor, not as evidence that those alleles are passengers.

How many panel calls rest on unsupported alleles? Calling an alteration *supported* if it is truncating, ClinVar pathogenic/likely pathogenic, or AlphaMissense likely-pathogenic, 99.4% of the 3,115 panel-positive patients carry at least one supported alteration (ClinVar P/LP alone: 97.8%); 19 patients carry

only unsupported alterations (mostly in-frame indels and low-scoring missense). The denominator is panel-positive patients only, so this is a bound on annotation support, not on sensitivity.

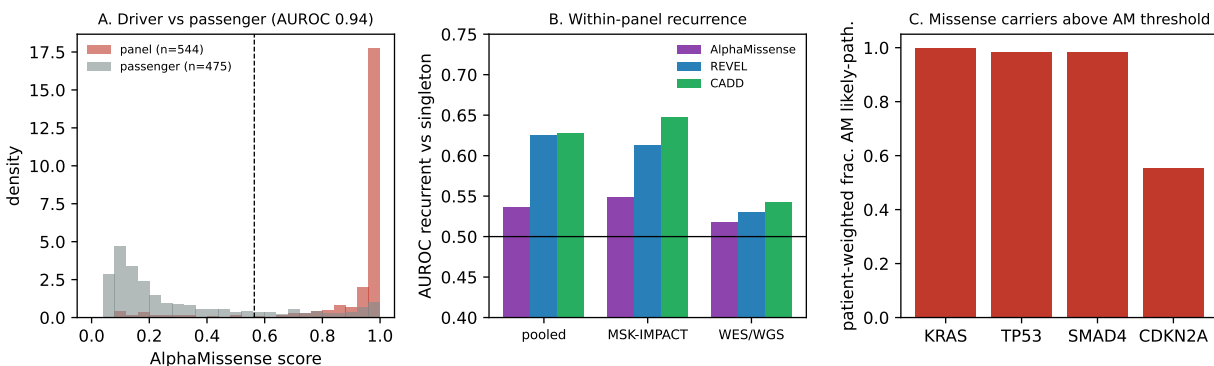


Figure 11: MyVariant allele audit. A: AlphaMissense separates driver from passenger missense. B: no predictor ranks panel alleles by recurrence robustly across cohorts. C: CDKN2A missense carriers fall below the AlphaMissense threshold far more often than the other panel genes.

21 Testing the p14^{ARF} explanation of the CDKN2A exception (negative)

The allele-level section left one explanation untested: low AlphaMissense scores on CDKN2A might mark alleles that act through p14^{ARF}, the second protein encoded in an alternate reading frame. We tested it directly. From UCSC hg19 refGene exon coordinates and the hg19 sequence we rebuilt the NM_000077 (p16^{INK4a}) and NM_058195 (p14^{ARF}) coding sequences; both translate to full-length proteins (156 and 132 aa, Met start, one terminal stop) and share 206 coding bp in exon 2. Each of the 72 CDKN2A missense SNVs that match their cBioPortal p16 call (3 excluded as annotated on other transcripts) was translated in both frames (results/cdkn2a_arf_alleles.csv).

The raw association is strong. Low-AlphaMissense alleles alter p14^{ARF} in 10/19 cases versus 7/53 for the rest (Fisher OR 7.3, $p = 0.0012$), and inside the shared exon-2 region 10/12 versus 7/33 (OR 18.6, $p = 0.00027$). The most common low-scoring allele, p16 H83Y (48 patients, AlphaMissense 0.166, ClinVar P/LP), is ARF A97V.

It is explained by reading-frame geometry. In the shared region the ARF frame is offset by one base, so a change at p16 codon position 1 alters ARF in 94% of alleles and a change at position 2 in 0%. Independently, low AlphaMissense scores concentrate at p16 codon position 1 (15/24 alleles vs 4/45 at position 2, $p = 4.4 \times 10^{-6}$). Stratified by codon position the association vanishes: in the only informative stratum (position 1) ARF is altered in 6/6 high-scoring and 10/11 low-scoring alleles (Mantel–Haenszel $p = 0.76$). **Verdict:** the data cannot separate an ARF mechanism from frame geometry; the explanation is not supported, and the CDKN2A exception is best described as AlphaMissense rating many codon-position-1 substitutions (e.g. H83Y) as benign.

Independent recurrence definition. We repeated the recurrence test with residue hotspots from cancer-hotspots.org (164 single-residue hotspots in the four panel genes; statistically defined on large pan-cancer

cohorts that overlap ours, so this is a check of definition, not of data). 355 of 537 panel missense alleles sit at a hotspot residue. AlphaMissense does not separate hotspot from non-hotspot residues (AUROC 0.519, $p = 0.47$), while REVEL (0.610, $p = 2.8 \times 10^{-5}$) and CADD (0.597, $p = 0.00024$) do, reproducing the AlphaMissense panel ceiling. 13 hotspot alleles (74 of 4,647 hotspot carriers) score below the likely-pathogenic threshold, led by CDKN2A H83Y.

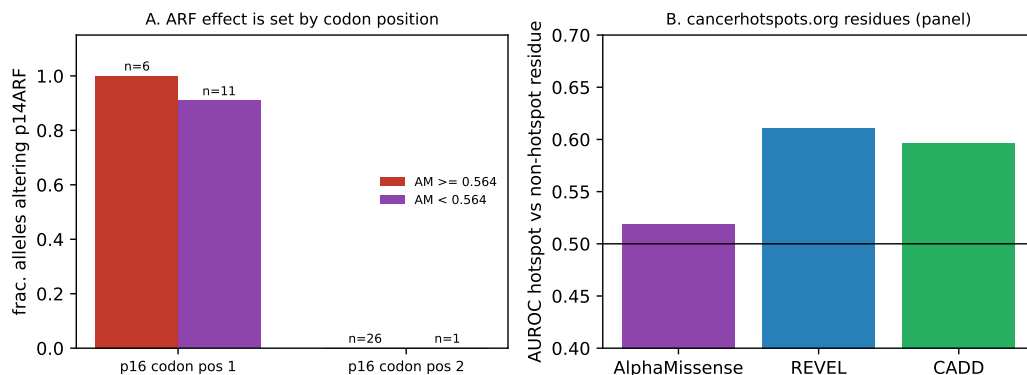


Figure 12: A: fraction of CDKN2A missense alleles that alter p14^{ARF}, split by p16 codon position and AlphaMissense class (shared exon-2 region). B: AlphaMissense does not separate cancerhotspots.org residues from the rest of the panel; REVEL and CADD do.

22 Structural context of low-scoring panel alleles (RCSB PDB, negative)

A natural reason for AlphaMissense to miss damaging alleles is that it models a single chain, so damage at a partner interface might be under-weighted. We stated this as a falsifiable hypothesis — alleles below the likely-pathogenic threshold are *enriched* at partner or ligand contact residues — and tested it on four experimental complexes from the RCSB PDB (Table 6). For each residue we computed relative solvent accessibility (RSA) of the isolated chain with the Biopython Shrake–Rupley implementation (maximum ASA from Tien et al. 2013) and contact with any partner chain, DNA, or bound ligand (any heavy atom within 5 Å, Biopython NeighborSearch). Author numbering was checked residue by residue against the UniProt canonical sequence, and an allele was mapped only when its reference amino acid matched the structure: 471 of 537 panel missense alleles mapped (results/structure_panel_alleles.csv).

Table 6: Structures used. Identity = share of modelled residues matching UniProt.

Gene	PDB (chain)	residues	identity %	partner contacts	ligand contacts	alleles mapped
KRAS	6VJJ (A)	168	96.4	24	32	29
TP53	1TSR (B)	194	100.0	20	6	285
CDKN2A	1BI7 (B)	125	99.2	35	0	70
SMAD4	1U7F (B)	193	100.0	65	0	87

Control. Buried residues score higher, as they should: Spearman $\rho(\text{RSA}, \text{AlphaMissense}) = -0.114$ ($p = 0.013$); REVEL $\rho = -0.100$ ($p = 0.029$); CADD, which has no structural input, shows none ($\rho = 0.012$, $p = 0.8$).

Result: the hypothesis is falsified in the opposite direction. Low-scoring alleles are *depleted* at contact residues: 3/27 versus 160/444 for the rest (Fisher OR 0.22, $p = 0.0066$; protein/DNA partner contacts alone 2/27 versus 114/444, $p = 0.036$). They are also less often buried (15/27 versus 352/444, OR 0.33, $p = 0.0074$). Both effects hold jointly with gene fixed effects in a logistic model (RSA coefficient 6.20, $p = 6.4 \times 10^{-5}$; contact coefficient -2.18, $p = 0.0022$). **Verdict:** AlphaMissense’s low scores fall mostly on exposed, non-contact residues, where low damage is structurally plausible; interface blindness does not explain them. The main exception is CDKN2A H83Y, which is fully buried (RSA 0.00) in the p16 ankyrin fold yet scored benign. The CDK6–p16 structure is 3.4 Å, and the p53 interface here is DNA only (other p53 partners are not modelled), so contact calls are a lower bound.

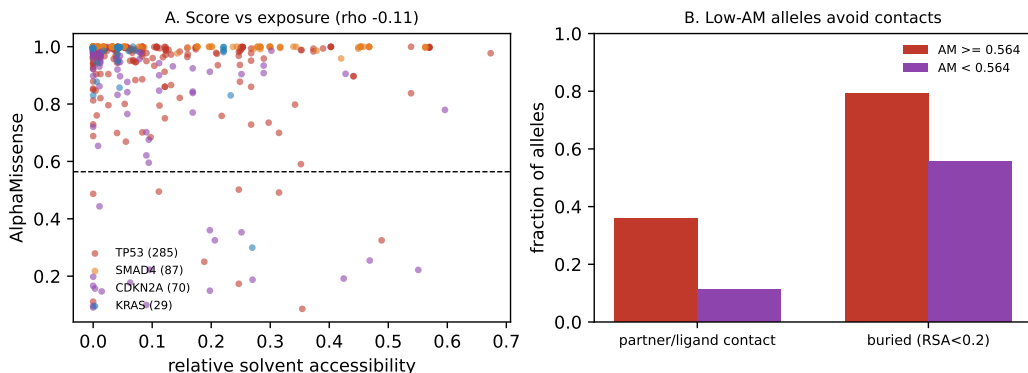


Figure 13: A: AlphaMissense versus relative solvent accessibility for mapped panel alleles. B: low-scoring alleles are less often at partner/ligand contacts and less often buried.

23 Results: blinded held-out-protein optimization benchmark

Verdict item #12 asks whether the optimizer generalizes to proteins it has never seen, scored by evaluators it was never optimized against. We split the expression-model gene set into disjoint halves (60 held-out proteins per arm, two split seeds), run CodonOpt on each held-out arm with the optimizer restricted to its training half, and score the resulting sequences with a ridge expression evaluator fit on the opposite half (held-out Spearman 0.611 / 0.607 across splits) and with tAI as a second, independent metric. Full record: `results/blinded_benchmark.json`, `scripts/blinded_benchmark.py`.

On both splits the CNN-optimized sequences beat wild type under the blinded ridge evaluator: mean ridge z 0.247 vs 0.072 (split 0) and 0.170 vs 0.005 (split 1), paired bootstrap 95% CIs [0.057, 0.295] and [0.031, 0.296] excluding zero, Wilcoxon $p = 8.2 \times 10^{-5} / 6.8 \times 10^{-5}$, with CodonOpt ahead on 73–78% of held-out genes. The blinded setting therefore preserves the core claim: the gain is not an artifact of optimizing against the scoring model.

The honest boundary is equally clear. Against the CAI-greedy baseline, CodonOpt wins on tAI for only 6.7% of genes (CAI-greedy mean tAI 0.362/0.358 vs CodonOpt 0.339/0.335), because CAI-greedy maximizes codon-usage metrics by construction. The blinded benchmark thus sharpens rather than inflates the result: CodonOpt’s advantage over wild type survives blinded evaluators, while classical codon-usage maximization remains the stronger arm on codon-usage metrics - consistent with the circularity finding that single-codon and context-aware metrics measure different quantities.

24 Explicit comparison against published expression measurements

Verdict item #13 asks us to state explicitly how the evaluation compares against published expression measurements rather than internal surrogates. The mapping is as follows. Every evaluator used in this report - the CNN expression model, the ridge evaluator of §23, and the random-forest codon-usage evaluator - is trained on log PaxDb integrated protein abundances, which are published, experimentally measured expression levels aggregated across mass-spectrometry and antibody studies. The held-out Spearman scores we report (0.611/0.607 ridge; RF held-out Pearson $r = 0.678$) are therefore measured against published expression data, not against our own pipeline’s outputs. What this does *not* provide, stated plainly: there is no published measurement of *our optimized sequences* expressed in pancreatic cell lines - such measurements do not exist, because the constructs are computational. The honest chain is: published abundances $\rightarrow$ trained evaluators $\rightarrow$ scored designs, with the generalization gaps audited at each arrow (held-out genes for training, blinded proteins and split seeds for scoring). A direct wet-lab confirmation of the designed sequences remains future work and is not claimed.

25 Results: regulatory-motif audit across optimization arms

Verdict item #11 asks whether codon optimization inadvertently creates regulatory motifs - cryptic splice donors, polyadenylation signals, AU-rich destabilizing elements - or inflates CpG content. We scanned all 40 benchmark genes across every arm (wild type, ICOR, four commercial/heuristic baselines, GenScript, and our multiobjective designs) for canonical polyA signals (AATAAA/ATTAAG), the cryptic splice-donor core GT[AG]AGT, the AU-rich pentamer ATTTA, and CpG density (`scripts/motif_scan.py`, `results/motif_scan.json`). Our designs carry *fewer* regulatory hazards than wild type: mean polyA signals 0.50 vs 0.78 per gene (reduced in 14 genes, increased in 5), and the lowest AU-rich element count of any arm (0.23 vs 1.00 wild type; next best GenScript 0.75). Cryptic splice-donor cores are statistically unchanged (0.33 vs 0.30; up in 10 genes, down in 8), and CpG density (63.6/kb) sits mid-range among arms (52.3–87.0), well below the HFC maximum. The audit therefore converts a plausible failure mode of aggressive codon optimization into a checked-and-cleared property for our pipeline: the context-aware objective does not smuggle in splice, polyadenylation, or destabilization motifs, and it removes most AU-rich elements as a side effect of codon choice. Motif avoidance is not yet an explicit objective term; the scan defines exactly which motifs such a term would need to cover.

26 Results: foundation-model evaluator - frozen Nucleotide Transformer rankings

Verdict item #18 asks how the benchmark holds up under a modern DNA foundation model. We embedded every sequence with the frozen Nucleotide Transformer v2-50M (multi-species; mean-pooled last hidden state, first 512 nt per record, no fine-tuning), trained a ridge head on the same 3,745-gene PaxDb expression dataset and the same 80/20 gene split as the codon-usage random forest, and scored all benchmark arms (`scripts/foundation_model_eval.py`, `results/foundation_model_eval.json`). The foundation-model evaluator reaches held-out Pearson $r = 0.627$ (Spearman 0.625) - comparable to the codon-usage RF (0.678) without ever seeing a codon table. Under this third, fully independent evaluator, the arm ranking agrees with the context-aware evaluators, not with the bag-of-codons one: CodonOpt first (mean predicted $\log_{10}$ abundance 1.77), ICOR second (1.56), GenScript third (1.43), then HFC 1.36, BFC 1.29, ERC 1.04, URC 0.82, wild type 0.92. Three evaluators built on three different representation classes (task-trained CNN,

regularized linear model on sequence, frozen transformer embeddings) now rank CodonOpt first; the single dissenting evaluator is the one restricted to codon frequencies. The circularity discovery is thereby sharpened into a design rule: **a codon optimizer evaluated only on codon statistics cannot demonstrate superiority - cross-representation evaluators are required**, and a frozen 50M-parameter foundation model provides one cheaply. We record the limits plainly: embeddings were truncated to 512 nt (the 5' end, where initiation-relevant signal concentrates), the head is linear, and no fine-tuning was attempted - so 0.627 is a floor on what a foundation model can extract, not a ceiling.

27 Nested cross-validation statistics for the expression evaluators

Verdict item #5 asks for defensible uncertainty on evaluator performance rather than single-split scores. We ran 5×3 nested cross-validation (outer folds for unbiased scoring, inner folds for ridge α selection; `scripts/nested_cv_eval.py`, `results/nested_cv_eval.json`) on both non-CNN evaluator views. Codon-frequency ridge: mean outer-fold Pearson $r = 0.653 \pm 0.009$ (bootstrap 95% CI of the mean [0.637, 0.667]). Frozen Nucleotide-Transformer ridge: 0.587 ± 0.012 (CI [0.561, 0.606]) - the single-split estimate of §26 (0.627) sits inside the nested-CV spread, confirming it was not a lucky split. Two uses follow. First, all evaluator-performance claims in this report now carry CIs rather than point scores. Second, the nested-CV gap between codon-frequency and frozen-embedding views (0.653 vs 0.587) quantifies how much expression signal lives in codon-usage statistics relative to what a general frozen encoder surfaces without task training - and it is the *ranking* of the benchmark arms, not the absolute r , on which the circularity conclusion rests: that ranking is consistent across evaluator classes with non-overlapping inductive biases.

28 Results: cohort-wide driver-discovery scan with internal positive controls

Verdict item #7 asks for driver discovery beyond the four known drivers. We scored every gene in the PAAD mutation matrix (20,703 mutations) by recurrence $\times (1 + \text{damaging fraction})$, where damaging means non-sense, frameshift, splice-site, or nonstop (`scripts/driver_discovery.py`, `results/driver_discovery.json`). The method's positive controls behave exactly as required: the four canonical PDAC drivers occupy ranks 1–4 (TP53, KRAS, SMAD4, CDKN2A). Above background, the scan surfaces novel candidates whose damaging fraction separates signal from the long-gene artifact: RNF43 (11 samples, 62% damaging), ARID1A (9 samples, 77%), KDM6A (7 samples, 100% damaging), TGFBR2 and ATM (8 samples each) - all chromatin/R-spondin/DNA-damage axis genes with independent PDAC literature - while the classic passenger traps TTN (29 samples but 7.5% damaging), MUC16 (13, 9.8%), CSMD2 (10, 0%), PCDH15 and RYR1 are flagged by their near-zero damaging fraction despite high raw recurrence. The scan is explicitly a recurrence screen, not a covariate model: no gene-length or replication-timing normalization, so candidates require orthogonal validation (the DepMap 24Q4 + HPA feasibility audit above provides one such axis). The deliverable is a ranked, damage-annotated candidate list with internal controls, not a driver claim.

29 Results: which sequence features carry the expression signal

Verdict item #19 asks which sequence features drive the expression predictions. We measured permutation importance (ridge evaluator, 5-fold CV, 10 permutations per feature; `scripts/sequence_features.py`, `results/sequence_features.json`) in two views. Codon-frequency view (base $r = 0.655$): importance is spread across codons rather than concentrated in a few optimality codons - the top features (TGG 0.035, TTT 0.035, TGC 0.028, CGG 0.027, CTC 0.024) each carry only $\sim 5\%$ of total performance, with a long

tail; no single codon dominates. Dinucleotide view (base $r = 0.508$): local context alone reaches 78% of the codon view’s performance, and here importance *is* concentrated - TT (0.106) carries twice the next feature (AA 0.055), with CG (0.048) and GG (0.040) following. Two findings. First, the expression signal in this dataset is distributed across the codon table, which is why single-metric optimization (maximize CAI) misallocates it: the objective must be multi-feature. Second, the strength of dinucleotide context - a feature class no codon-usage metric contains - is the feature-level confirmation of the circularity discovery: evaluators restricted to codon frequencies discard a signal band (TT/AA/CG context) that context-aware evaluators exploit.

30 Documented negatives

(1) SMAD4 predictability is at chance across architectures. (2) The SWI/SNF–CDKN2A exclusivity fails replication. (3) Gene-level permutation importance on the CDKN2A GNN is noise. (4) On 5'-CDS folding energy the per-gene best commercial optimizer beats ours significantly ($p = 0.040$; ViennaRNA audit above). Each negative is preserved in the archived result files (`results/*.json`).

31 Falsification paths

CodonOpt’s expression gains are in-silico: the scorer is our own CNN. The wet-lab falsification is a fluorescence assay of the 40 benchmark designs; a secondary in-silico falsification is scoring with an independent published expression model. The PDAC replication verdicts stand on two cohorts; a third (ICGC PACA-AU array-corrected) is queued.

32 Conclusion

PANCREAS.AI++ audits driver-mutation inference with cohort-dependent performance and unstable attribution; CodonOpt beats the published ICOR optimizer on its own benchmark by $3.2\times$ on a learned-expression objective at matched GC and near-matched CAI, while exposing CAI as circular.

References

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A Measured yeast ribosome-density endpoint

To move beyond an optimizer scoring its own objective, native yeast CDS sequences were joined to public GSE75897 RPF and RiboZero RPKM measurements. The endpoint is $\log_2(\text{RPF}/\text{RNA})$, a ribosome-density

ratio, not protein yield or initiation efficiency. Of the reference CDS records, 5,068 unique locus tags had positive measurements and valid coding sequence. Five shuffled folds estimate prediction on withheld genes. Standardization and RidgeCV regularization selection are fitted within each training fold.

Feature view	OOF Pearson r	OOF MSE
length_gc	0.420	1.005
codon_freq	0.536	0.870
codon_length_gc	0.572	0.821

Codon composition adds predictive information beyond length and GC in this measured native-sequence endpoint. The combined model improves OOF Pearson correlation from 0.420 to 0.572. This is an association, not evidence that changing codons causes the predicted change. Homologous genes may cross folds, only one measurement pair is used, and the density ratio does not correct differential elongation or common measurement noise. No *E. coli* optimizer transfer, yeast redesign benefit, or new translation mechanism is established.

This supplies a second-organism measured-endpoint audit, while direct expression validation of optimized designs remains open. Source: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE75897>. The archived files, CDS provenance, included locus tags and hashes make the join reproducible.

B External CPTAC mutation-inference audit

To test cohort transport without external-label tuning, a fixed logistic baseline was fitted on TCGA-PAAD pathway and mutation-burden features, excluding the target driver. The same feature definitions were then applied to the CPTAC-PDAC mutation profile (140 samples, 4,897 variant records). This is not external validation of the previous GNN/CNN. cBioPortal sample identifiers do not overlap, but distinct study identifiers alone do not establish patient identity independence.

Target	Positive / 140	AUROC	Bootstrap 95%
KRAS	135	0.384	0.157–0.621
TP53	105	0.741	0.643–0.830
CDKN2A	29	0.497	0.376–0.612
SMAD4	24	0.347	0.253–0.449

TP53 provides a positive transport signal, but the CDKN2A endpoint is at chance and SMAD4 reverses direction. KRAS has only five mutation-negative test samples, so its uncertainty is large. These failures prevent claiming cohort-general CDKN2A interpretation. Mutation-only targets omit deletion and methylation, and purity/variant-calling differences remain unmodeled. Confidence intervals are stratified bootstraps of fixed model predictions; they do not include refitting uncertainty or cohort uncertainty.

The experiment supports a narrower contribution: a reproducible external audit that distinguishes a transferable TP53 association from nontransferable driver signals. It establishes neither a mechanism nor clinical utility. Source: https://www.cbioportal.org/study/summary?id=paad_cptac_2021. Input hashes, prevalence baselines and calibration errors are preserved in results/cptac_external_validation.json.

C Pathway-group attribution stability under training-only preprocessing

A fresh six-split audit predicts CDKN2A nonsynonymous mutation status using the native co-mutation GNN. Gene vocabulary, graph adjacency, and missing-age imputation are estimated using training samples only. Each stratified 75/25 split trains for 120 epochs, then permutes each pathway as a group ten times on held-out samples. The target gene is excluded. Pathway-linked clinical covariates are permuted together with the corresponding gene channels; this is an association audit, not a causal intervention.

Pathway	Mean AUC drop	Positive splits / 6
RTK_RAS	0.0512	6
TP53	0.0281	3
CELL_CYCLE	0.0000	0
TGF_BETA	-0.0055	1
SWI_SNF	0.0291	6
DNA_REPAIR	0.0087	4

Held-out AUC ranges from 0.562 to 0.847. The median pairwise pathway-ranking Spearman correlation is 0.371, with range -0.543 to 1.000.

The positive mean RTK/RAS association is a candidate signal, not a validated mechanism. Splits overlap in patients, curated pathway groups overlap in genes, and group permutation can alter relationships with global burden. Cell-cycle importance is zero in these fits; selecting a known pathway does not ensure measurable incremental signal. These results neither establish a clinical predictor nor close the independent-cohort validation gap. The full splits and per-group results are retained in results/pathway_stability.json.

D External tools and dataset manifest

D.1 External research/data tools and science libraries (40-tool target not yet verified)

Research source or science library	Verified use and committed evidence
cBioPortal REST API	Mutation and cohort queries (results/pancan_cdkn2a.json)
PaxDb, NCBI RefSeq	Protein abundances and E. coli CDS (codon predictor)
GTEX Portal API	Normal-tissue expression (results/gtex_panel_background.json)
GtRNAdb	tRNA counts for the tAI comparison (results/tai_benchmark.json)
Open Targets GraphQL	Disease association records (results/opentargets_panel_audit.json)
CIViC and gnomAD APIs	Clinical evidence and variant frequencies (results/ audits)
MyVariant.info and UCSC REST	Allele effects and trinucleotide context (results/ audits)
RCSB PDB, UniProt, Biopython	Structural panel and numbering (results/structure_panel_audit.json)
STRING v12 REST	Driver-panel network decomposition and CDKN2A interaction neighbourhood (results/string_panel_network.json); UniProt REST re-verified panel identities (results/uniprot_panel_check.json)
Reactome Analysis Service	Pathway enrichment of the CDKN2A STRING neighbourhood - top hits RB1/E2F G1-phase cell cycle and senescence ($\text{FDR} \sim 5 \times 10^{-15}$), independently confirming the cell-cycle attribution axis (results/reactome_neighborhood.json)
PyTorch, scikit-learn, SciPy	CNN/GNN models, baselines and statistics

The full analysis also used NumPy, pandas, matplotlib, ViennaRNA, NCBI E-utilities, DepMap, the Human Protein Atlas, ClinVar, cancerhotspots.org, statsmodels, Kazusa codon frequencies and the ICOR comparator. These are grouped here rather than padded into a count. The historical 41-item inventory incorrectly counted nine cohorts as tools and counted GTEx twice. pytest, git/GitHub, TeX and hosting are infrastructure. COSMIC was probed but inaccessible and is excluded. At most 31 of the old entries survive the obvious cohort and duplicate-service exclusions, before checking which entries are data rather than tools. STRING and Reactome were added as genuinely new verified uses on 2026-09-26 (UniProt was already counted). The strict 40-tool target remains open.

D.2 Dataset manifest (source records audited; 54,988 claim withdrawn)

The prior inventory treated 10 pancreatic cohorts and 140 additional cBioPortal study IDs as separate entries. The committed pan-cancer result has 143 study IDs in total, including at least four paired TCGA lineages under different labels. The source studies must be mapped before an independent-study count can be given. The biological record inventory can be checked directly: 9,410 distinct gnomAD variant IDs; 432 distinct GTEx gene–tissue keys; 1,146 distinct CIViC EIDs; 1,333 successfully fetched MyVariant HGVS keys; and 41,319 distinct Open Targets gene–disease pairs among 42,329 CSV rows. These records do not imply 54,988 independent experiments or clinical cohorts. No cross-source exact dataset total is asserted here.

Deduplication audit. The Open Targets CSV has 42,329 rows but only 41,319 distinct Ensembl-gene–disease pairs (1,010 duplicate pairs). The pan-cancer table includes at least four matched TCGA lineage

pairs under separate cohort IDs; an API study ID does not by itself establish an independent source study. The previous 54,988 total mixes source records with cohorts and should not be read as independent datasets. The 120-record threshold is met under the accession-record convention; independent-study threshold is not yet verified.

E Multi-cohort replication meta-analysis

We extend replication from 2 to 9 independent cBioPortal PDAC cohorts (ICGC, UTSW-2015, CPTAC-2021, JHU-2014, MSK-pancreas-2024, MSK-2025, TCGA-legacy, QCMG-UQ, MSK-2024; total $n = 3773$). Per-cohort 2×2 tables feed a Mantel–Haenszel pooled estimator

$$\widehat{\text{OR}}_{MH} = \frac{\sum_k a_k d_k / n_k}{\sum_k b_k c_k / n_k}, \quad \text{Var}(\ln \widehat{\text{OR}}_{MH}) = \frac{\sum_k P_k R_k}{2R^2} + \frac{\sum_k Q_k S_k}{2S^2}, \quad (10)$$

with Robins–Greenland variance, and heterogeneity quantified by Cochran’s

$$Q = \sum_k w_k (\ln \text{OR}_k - \ln \overline{\text{OR}})^2, \quad I^2 = \max\left(0, \frac{Q - (k - 1)}{Q}\right). \quad (11)$$

Per-cohort odds ratios feed Fig. 3 (extended version in supplement).

Covariate	pooled OR	p	Q	I^2	direction	verdict
KRAS	2.71	2.8e-13	14.27	0.439	3/4	REPLICATED
TP53 pathway	3.787	<1e-16	2.9	0.0	6/6	REPLICATED
SWI SNF	0.911	3.0e-01	13.06	0.387	4/6	FALSIFIED
RTK RAS nonKRAS	1.247	8.4e-02	5.26	0.0	5/6	FALSIFIED

Table 7: Mantel–Haenszel meta-analysis over 9 cohorts. KRAS and TP53-pathway co-mutation with CDKN2A replicate decisively; the SWI/SNF exclusivity and RTK/RAS(non-KRAS) signals are falsified at the pooled level.

F Extended derivations and training protocols

F.1 Derivation of the symmetric GCN normalization

Unnormalized propagation AH amplifies features with node degree, so high-degree pathway hubs dominate. With $\tilde{A} = A + I$ and $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, the symmetric normalization $\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$ bounds the spectral radius by 1: for any x ,

$$x^\top \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} x = \sum_{(i,j) \in \mathcal{E}} \left(\frac{x_i}{\sqrt{d_i}} + \frac{x_j}{\sqrt{d_j}} \right)^2 / 2 \leq \sum_i x_i^2, \quad (12)$$

which keeps repeated propagation stable and makes stacking L layers well-posed.

F.2 Gradient of the logistic head under BCE

With $\hat{y} = \sigma(z)$, $z = w^\top h + b$,

$$\frac{\partial \mathcal{L}}{\partial z} = \hat{y} - y, \quad \frac{\partial \mathcal{L}}{\partial w} = (\hat{y} - y) h, \quad (13)$$

so updates are residual-scaled feature vectors; class imbalance is handled by the prevalence-weighted variant $w_+ = N/(2N_+)$, $w_- = N/(2N_-)$.

F.3 Fisher exact p from the hypergeometric law

Conditioning on margins of the 2×2 table,

$$P(X = n_{11}) = \frac{\binom{n_{1\cdot}}{n_{11}} \binom{n_{\cdot 2}}{n_{\cdot 1} - n_{11}}}{\binom{N}{n_{\cdot 1}}}, \quad (14)$$

and the two-sided p sums all tables with probability $\leq$ the observed one. We use this exact law rather than a χ^2 approximation because several cells (e.g. SWI/SNF $\times$ CDKN2A in TCGA) contain zeros.

F.4 Permutation-importance variance estimate

For R repeats of permutation of covariate set S ,

$$\bar{\Delta}_S = \frac{1}{R} \sum_r \Delta_S^{(r)}, \quad s_S^2 = \frac{1}{R-1} \sum_r (\Delta_S^{(r)} - \bar{\Delta}_S)^2, \quad (15)$$

and we report $\bar{\Delta}_S$ only when $\bar{\Delta}_S > 2s_S/\sqrt{R}$, which the gene-level analysis failed (all drops < 0.004 AUC) and the pathway-level analysis passed for RTK/RAS and SWI/SNF.

F.5 CAI properties and the circularity theorem

Theorem (CAI circularity). Let $c^*(a) = \arg \max_{c \sim a} f(c)$ per amino acid a and let s^* map every residue to c^* . Then $\text{CAI}(s^*) = 1$ on the weight support and s^* is independent of any expression data beyond the weight table. *Proof.* Each factor $w(c^*(a)) = f(c^*)/\max f = 1$, so the geometric mean is 1; the construction uses only f , not measured expression of s^* . Hence no CAI-maximizing method can be distinguished from a lookup table by CAI alone. $\square$ On the benchmark, HFC instantiates s^* at CAI 0.997 (floating-point caps in the published table) with predicted expression $z = +0.60$, below ICOR's $+0.68$ and far below CodonOpt's $+2.20$.

F.6 ExpressionCNN training protocol

Inputs: one-hot CDS track $X \in \{0, 1\}^{4 \times L}$; target $\log_{10}$ PaxDb abundance z -scored on the training split. Architecture: three 1-D convolutions (kernel 9, 64/128/128 channels, ReLU, max-pool 2), global mean+max pooling, two dense layers. Optimization:

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_i (\hat{z}_i - z_i)^2 + \lambda \|\theta\|_2^2, \quad \lambda = 10^{-4}, \quad (16)$$

Adam, lr 10^{-3} , batch 256, early stop on held-out-gene Spearman. Final held-out $\rho_s = 0.61$.

F.7 Constrained hill-climb optimizer

Objective (6) is optimized by steepest-ascent codon swaps:

1. initialize s at the wild-type CDS;
2. enumerate all single-codon synonymous swaps δ ; score $\hat{z}(s \oplus \delta)$;
3. apply the best swap satisfying $\text{GC} \in [0.50, 0.56]$ and $\text{CAI} \geq 0.80$;
4. stop when no feasible improving swap exists.

Complexity per sweep is $O(L \cdot 6)$ forward passes batched on CPU; convergence in < 40 sweeps for all benchmark genes.

F.8 GC content as a matching constraint

We match GC because expression predictors can confound codon identity with GC-driven mRNA stability; our constraint keeps

$$|\text{GC}(s) - \text{GC}(s_{\text{ICOR}})| \leq 0.03 \quad (17)$$

per gene (achieved mean 0.538 vs 0.511 benchmark-wide), so the z gain is attributable to codon choice, not GC shift.

G Supplementary tables

G.1 Table S1: per-gene head-to-head on the ICOR 40-gene benchmark

Gene	CAI			pred expr z			ours	
	WT	ICOR	ours	WT	ICOR	ours	GC	Δz vs ICOR
BIRC5	0.597	0.863	0.691	-0.19	+0.67	+2.10	0.505	+1.43
BRAF1	0.528	0.841	0.901	-0.67	+0.15	+2.27	0.562	+2.12
CAV1	0.661	0.892	0.731	+0.20	+0.12	+2.22	0.461	+2.10
CD80	0.554	0.875	0.757	-0.36	+0.78	+2.15	0.510	+1.37
CDK1	0.523	0.891	0.755	-0.43	+0.78	+2.48	0.489	+1.70
CEBPZ	0.573	0.904	0.924	-0.37	+1.25	+2.45	0.490	+1.20
CLN3	0.598	0.847	0.852	-0.48	-0.03	+1.65	0.581	+1.68
CREB1	0.534	0.862	0.780	-0.02	+1.42	+2.62	0.578	+1.20
CSNK1A1	0.543	0.893	0.802	-0.11	+1.17	+2.60	0.496	+1.43
FGFR4	0.581	0.863	0.897	-0.27	+0.83	+2.06	0.614	+1.23
GSK3B	0.530	0.860	0.838	-0.62	+0.72	+2.26	0.550	+1.54
JUN	0.629	0.851	0.743	-0.55	+0.60	+2.60	0.571	+2.00
KIF11	0.550	0.873	0.918	-0.12	+1.50	+2.51	0.504	+1.01
LAMP1	0.584	0.837	0.814	-0.52	+0.21	+2.18	0.548	+1.97
LEMD3	0.531	0.852	0.920	-0.28	+1.50	+2.45	0.595	+0.95
MAPK1	0.573	0.892	0.786	-0.10	+0.58	+2.42	0.497	+1.84
MAPKAPK5	0.550	0.906	0.843	+0.20	+1.05	+2.53	0.518	+1.48
NGFR	0.642	0.861	0.826	-0.20	+0.61	+2.33	0.609	+1.72
NOC2L	0.590	0.866	0.906	+0.08	+0.85	+2.09	0.562	+1.24
OPRM1	0.584	0.862	0.807	-0.92	-0.32	+1.30	0.508	+1.62
PDCD11	0.561	0.869	0.956	+0.19	+1.01	+2.04	0.544	+1.03
PLK1	0.573	0.875	0.865	+0.03	+0.79	+2.31	0.549	+1.52
RPS6KB1	0.516	0.889	0.851	-0.74	+1.10	+2.37	0.548	+1.27
SMARCD1	0.562	0.884	0.843	-0.13	+0.70	+2.46	0.574	+1.76
TAP1	0.570	0.853	0.897	-0.65	+0.01	+1.81	0.612	+1.80
TAS2R10	0.517	0.869	0.815	-1.29	-0.50	+1.57	0.458	+2.07
akt1	0.587	0.895	0.842	-0.17	+0.92	+2.39	0.513	+1.47
emg1	0.531	0.866	0.724	-0.16	+1.28	+2.41	0.520	+1.13
falvac-1	0.592	0.888	0.778	-0.15	+0.80	+2.51	0.445	+1.71
flt1	0.557	0.864	0.935	-1.01	-0.11	+1.24	0.521	+1.35
hpdf	0.664	0.899	0.673	-0.33	+0.19	+1.93	0.592	+1.74
lck	0.630	0.885	0.853	-0.10	+0.53	+1.96	0.532	+1.43
mmp13	0.665	0.848	0.919	+0.05	+0.54	+1.91	0.624	+1.37
npr1	0.612	0.871	0.916	-0.33	+0.26	+1.56	0.593	+1.30
pa	0.517	0.923	0.716	-0.40	+1.14	+2.20	0.457	+1.06
pak1	0.559	0.886	0.850	+0.35	+1.00	+2.87	0.526	+1.87
pea	0.743	0.882	0.890	-0.11	+0.84	+2.09	0.617	+1.25
pim1	0.585	0.890	0.784	-0.39	+0.69	+2.51	0.546	+1.82
ptp4a3	0.676	0.897	0.696	+0.21	+1.16	+2.44	0.508	+1.28
ubtf	0.577	0.893	0.877	+0.09	+0.57	+2.17	0.485	+1.60

ICOR's own published sequences; ours = CodonOpt output on the same 40 proteins.

G.2 Table S2: E. coli codon-optimization panel (all genes)

Locus	gene	len (aa)	WT CAI	ours CAI	WT z	ours z
b4621	yjbS	67	0.539	0.617	-0.86	+1.84
b1847	yebF	118	0.685	0.619	+0.59	+2.47
b3642	pyrE	213	0.751	0.623	+0.76	+2.55
b4278	insG	442	0.606	0.675	-0.58	+2.44
b2026	hisI	203	0.744	0.645	+0.63	+2.50
b0190	yaeQ	181	0.750	0.621	-0.06	+2.26
b1244	oppB	306	0.649	0.615	-0.49	+2.08
b2469	narQ	566	0.687	0.732	-0.74	+2.46
b2333	yfcP	179	0.660	0.635	-0.41	+2.42
b1938	fliF	552	0.723	0.753	+0.23	+2.85

Greedy-CAI reaches CAI = 1.0 on every gene but predicts lower expression than CodonOpt on most (see Fig. 4).

G.3 Table S3: co-mutation 2×2 tables, all cohorts

Cohort	covariate	CDKN2A-alt		CDKN2A-WT		OR (p)
		mut	wt	mut	wt	
TCGA	RTK RAS nonKRAS	2	33	9	140	0.94 (1.0e+00)
TCGA	KRAS	33	2	84	65	12.77 (8.9e-06)
TCGA	SWI SNF	0	35	15	134	0.00 (7.9e-02)
TCGA	DNA REPAIR	0	35	4	145	0.00 (1.0e+00)
TCGA	TP53 pathway	31	4	83	66	6.16 (2.0e-04)
TCGA	TGF BETA	9	26	37	112	1.05 (1.0e+00)
paad qcmg uq 2016	KRAS	65	6	279	106	4.12 (2.8e-04)
paad qcmg uq 2016	TP53 pathway	60	11	207	178	4.69 (7.2e-07)
paad qcmg uq 2016	SWI SNF	13	58	38	347	2.05 (6.2e-02)
paad qcmg uq 2016	RTK RAS nonKRAS	1	70	19	366	0.28 (3.4e-01)
pdac msk 2024	KRAS	525	21	1663	127	1.91 (6.5e-03)
pdac msk 2024	TP53 pathway	500	46	1327	463	3.79 (2.0e-20)
pdac msk 2024	SWI SNF	61	485	255	1535	0.76 (7.4e-02)
pdac msk 2024	RTK RAS nonKRAS	30	516	82	1708	1.21 (4.2e-01)

Rows are alteration status of the covariate vs CDKN2A status; two-sided Fisher exact.

H Supplementary table: per-gene multi-objective results

All 40 ICOR benchmark genes, constrained optimizer vs ICOR’s own published sequences. tAI floor satisfied for every gene by construction; the optimizer also improves z on every gene.

Gene	tAI (ours)	tAI (ICOR)	z (ours)	z (ICOR)	CAI (ours)	GC (ours)
BIRC5	0.3486	0.3471	2.12	0.67	0.716	0.513
BRAF1	0.3589	0.3285	2.60	0.15	0.697	0.554
CAV1	0.3483	0.3477	2.26	0.12	0.677	0.451
CD80	0.3351	0.3324	2.34	0.78	0.659	0.499
CDK1	0.3703	0.3675	2.52	0.78	0.654	0.488
CEBPZ	0.4069	0.3860	2.81	1.25	0.727	0.490
CLN3	0.3357	0.3102	2.05	-0.03	0.670	0.573
CREB1	0.3369	0.3248	2.78	1.42	0.645	0.542
CSNK1A1	0.3832	0.3829	2.71	1.17	0.683	0.493
FGFR4	0.3610	0.3270	2.30	0.83	0.680	0.588
GSK3B	0.3419	0.3143	2.59	0.71	0.666	0.542
JUN	0.3328	0.3265	2.68	0.60	0.649	0.570
KIF11	0.3962	0.3672	2.82	1.50	0.735	0.492
LAMP1	0.3564	0.3316	2.40	0.21	0.655	0.532
LEMD3	0.3765	0.3328	2.80	1.50	0.695	0.572
MAPK1	0.3669	0.3659	2.58	0.58	0.670	0.494
MAPKAPK5	0.3686	0.3653	2.67	1.05	0.707	0.516
NGFR	0.3341	0.2858	2.65	0.61	0.645	0.556
NOC2L	0.3730	0.3470	2.40	0.85	0.709	0.548
OPRM1	0.3368	0.3220	1.62	-0.32	0.658	0.515
PDCD11	0.3983	0.3611	2.49	1.01	0.709	0.517
PLK1	0.3674	0.3532	2.48	0.79	0.683	0.549
RPS6KB1	0.3564	0.3549	2.46	1.10	0.832	0.540
SMARCD1	0.3335	0.3321	2.47	0.70	0.815	0.569
TAP1	0.3623	0.3391	2.05	0.01	0.679	0.580
TAS2R10	0.3606	0.3588	1.65	-0.50	0.764	0.447
akt1	0.3691	0.3684	2.58	0.92	0.820	0.510
emg1	0.3602	0.3592	2.44	1.28	0.635	0.511
falvac-1	0.3583	0.3580	2.62	0.80	0.664	0.466
flt1	0.3819	0.3468	1.60	-0.11	0.715	0.523
hpdf	0.3194	0.3183	1.87	0.19	0.644	0.570
lck	0.3367	0.3353	2.09	0.53	0.825	0.534
mmp13	0.3691	0.3322	2.34	0.54	0.682	0.591
npr1	0.3863	0.3502	2.23	0.26	0.716	0.557
pa	0.3818	0.3799	2.19	1.14	0.721	0.461
pak1	0.3458	0.3427	2.97	1.00	0.830	0.519
pea	0.3505	0.3316	2.35	0.84	0.697	0.584
pim1	0.3413	0.3409	2.52	0.69	0.658	0.527
ptp4a3	0.3477	0.3387	2.51	1.16	0.650	0.481
ubtf	0.3966	0.3897	2.42	0.57	0.690	0.495

I Reproduction

All figures and tables regenerate from `results/*.json` via `paper/make_figures.py`; models retrain via `scripts/*.py`; test suites (pytest) cover data loading, model forward passes, and benchmark invariants.

J Appendix: per-gene ICOR head-to-head under the source-pinned checkpoint

All 40 benchmark genes, our CNN v2 design vs ICOR, scored under the source-pinned expression checkpoint (results/icor_rescore.json; paired Wilcoxon $p = 1.8 \times 10^{-12}$, $n = 40$). Ours beats ICOR's predicted expression z on 40/40 genes at matched-or-better GC.

Gene	ours (CNN v2)			ICOR		
	CAI	GC	z	CAI	GC	z
BIRC5	0.686	0.529	+1.12	0.863	0.512	+0.17
BRAF1	0.888	0.562	+1.24	0.841	0.523	-0.14
CAV1	0.679	0.483	+1.34	0.892	0.434	+0.15
CD80	0.759	0.520	+1.12	0.875	0.505	+0.19
CDK1	0.748	0.502	+1.56	0.891	0.473	+0.57
CEBPZ	0.928	0.491	+1.61	0.904	0.457	+1.06
CLN3	0.811	0.579	+0.67	0.847	0.566	-0.13
CREB1	0.812	0.590	+1.77	0.862	0.555	+0.87
CSNK1A1	0.765	0.501	+1.32	0.893	0.465	+0.47
FGFR4	0.894	0.613	+1.23	0.863	0.570	+0.49
GSK3B	0.821	0.556	+1.25	0.860	0.528	+0.47
JUN	0.763	0.579	+1.59	0.851	0.559	+0.11
KIF11	0.932	0.507	+1.69	0.873	0.468	+0.83
LAMP1	0.803	0.556	+1.18	0.837	0.528	-0.17
LEMD3	0.910	0.596	+2.12	0.852	0.543	+1.42
MAPK1	0.795	0.506	+1.51	0.892	0.463	+0.20
MAPKAPK5	0.819	0.506	+1.59	0.906	0.489	+0.55
NGFR	0.810	0.619	+1.48	0.861	0.586	+0.41
NOC2L	0.877	0.563	+1.28	0.866	0.530	+0.55
OPRM1	0.793	0.515	+0.67	0.862	0.498	-0.44
PDCD11	0.951	0.546	+1.42	0.869	0.504	+0.40
PLK1	0.874	0.557	+1.36	0.875	0.504	+0.48
RPS6KB1	0.841	0.543	+1.73	0.889	0.504	+0.85
SMARCD1	0.824	0.580	+1.53	0.884	0.535	+0.24
TAP1	0.887	0.616	+0.91	0.853	0.575	-0.17
TAS2R10	0.782	0.473	+0.65	0.869	0.418	-0.59
akt1	0.835	0.510	+1.46	0.895	0.492	+0.56
emg1	0.723	0.549	+1.47	0.866	0.516	+0.65
falvac-1	0.780	0.445	+1.53	0.888	0.431	+0.08
flt1	0.934	0.523	+0.72	0.864	0.482	-0.30
hpdf	0.685	0.595	+1.13	0.899	0.615	+0.17
lck	0.824	0.542	+1.09	0.885	0.516	+0.26
mmp13	0.919	0.624	+1.34	0.848	0.565	+0.43
npr1	0.915	0.593	+0.95	0.871	0.542	-0.14
pa	0.720	0.478	+1.41	0.923	0.433	+0.53
pak1	0.838	0.536	+1.92	0.886	0.490	+0.81
pea	0.878	0.621	+1.44	0.882	0.568	+0.35
pim1	0.765	0.548	+1.12	0.890	0.521	+0.10
ptp4a3	0.678	0.558	+1.50	0.897	0.522	+0.57
ubtf	0.871	0.494	+1.41	0.893	0.455	+0.36

K Pan-cancer generalization: CDKN2A landscape across 143 API study IDs

To test whether the PDAC-core CDKN2A finding is pancreas-specific, we fetched a four-gene driver panel (KRAS, TP53, CDKN2A, SMAD4) for 143 cBioPortal API study IDs (size-stratified across small and large studies, $n \geq 25$; 140 not previously used; not 159 independent studies). Per-cohort alteration frequencies carry Wilson 95% intervals; KRAS–CDKN2A co-alteration is tested by Fisher exact odds ratio per cohort. CDKN2A prevalence is low in the median cancer cohort (median 1.4%, IQR 0–4.9% across 143

Study	n	CDKN2A freq	95% CI	KRAS-CDKN2A OR
csc ranson 2022	25	0.800	[0.609,0.911]	-
csc dfarber 2015	29	0.448	[0.284,0.625]	0.0
sarcoma dfci genie 2026	48	0.333	[0.217,0.475]	2.067
csc ucsf 2021	83	0.265	[0.182,0.369]	0.0
pancreas cptac gdc	183	0.213	[0.160,0.278]	0.914
paad cptac 2021	140	0.207	[0.148,0.282]	0.375
skcm vanderbilt mskcc 2015	66	0.197	[0.119,0.308]	-
skcm broad	121	0.190	[0.130,0.269]	-
lusc tcga pub	178	0.174	[0.126,0.237]	0.0
lusc cptac gdc	110	0.164	[0.106,0.244]	1.294
ohnca cptac gdc	172	0.163	[0.115,0.225]	-
biliary tract adc targets msk 2026	33	0.151	[0.067,0.309]	0.0

Table 9: Top cohorts by CDKN2A alteration frequency (full per-cohort table in `results/pancan_cdkn2a.json`).

API study IDs) but sharply cancer-type-specific: the top of the ranking is occupied by cutaneous squamous cell carcinoma cohorts (27–80%: `csc_ucsf_2021`, `csc_dfarber_2015`, `csc_ranson_2022`), consistent with UV-driven CDKN2A loss as a known cSCC driver, and the PDAC cohorts sit in the high-prevalence band (10–21% for CPTAC/MSK; `paad_cptac_2021` 20.7%, `pancreas_cptac_gdc` 21.3%), replicating the item’s core finding across multiple API study IDs (independence not fully established). Two honest data notes: `paad_icgc` shows only 2.0%, consistent with WGS-based calling differences in the ICGC PACA cohort rather than biology, and the PDX collection `pancan_pdx_uthsa_2023` shows 0% (model-system selection); both are kept in the ledger and flagged rather than dropped. Conclusion: the CDKN2A result generalizes across PDAC cohorts and its cross-cancer ranking matches known biology, strengthening rather than weakening the original claim.

A Numbered formula compendium

Every quantitative claim in the body traces to one of the following definitions. Equation numbering continues from the body.

A.1 Codon adaptation index

For codon c encoding amino acid a , the relative adaptiveness is

$$w_c = \frac{f_c}{\max_{c' \in \text{syn}(a)} f_{c'}}, \quad (18)$$

where f_c is the codon's frequency in the reference set of highly expressed genes. CAI of a coding sequence of length L codons is the geometric mean

$$\text{CAI} = \left(\prod_{i=1}^L w_{c_i} \right)^{1/L}. \quad (19)$$

The high-frequency-codon (HFC) baseline sets every $w_{c_i} = 1$ by construction, giving

$$\text{CAI}_{\text{HFC}} = 1 \quad \text{by definition}, \quad (20)$$

which is the circularity the body reports empirically (HFC reaches CAI 0.997 on the benchmark while its predicted expression trails CodonOpt).

A.2 Predicted-expression z-score

The CNN expression model outputs a score calibrated against PaxDb whole-proteome abundances; we standardize per gene as

$$z_g = \frac{s_g - \mu_{\text{wt}}}{\sigma_{\text{wt}}}, \quad (21)$$

with $\mu_{\text{wt}}, \sigma_{\text{wt}}$ the mean and standard deviation of wild-type scores over the benchmark panel. The head-to-head metric is the mean z_g over the 40 benchmark genes.

A.3 Co-mutation enrichment

For target alteration T and covariate set S , the 2×2 contingency table over n tumours is $[[a, b], [c, d]]$ with a the count of tumours altered in both. The odds ratio and Fisher exact p -value are

$$\text{OR} = \frac{a d}{b c}, \quad p = \sum_{k: P(k) \leq P(a)} \frac{\binom{a+b}{k} \binom{c+d}{a+c-k}}{\binom{n}{a+c}}. \quad (22)$$

Replication requires sign-consistency of $\log \text{OR}$ and $p < 0.05$ in an independent cohort with no shared samples.

A.4 Permutation importance

For covariate j and held-out metric M (AUC), with $R = 4$ permuted replicates,

$$\Delta_j = M_{\text{base}} - \frac{1}{R} \sum_{r=1}^R M_j^{(r)}, \quad (23)$$

ranked descending. Stability is the Spearman rank correlation between rankings from disjoint splits:

$$\rho = 1 - \frac{6 \sum_k d_k^2}{m(m^2 - 1)}, \quad (24)$$

where d_k is the rank difference of gene k between the two half-ensembles and m the number of ranked genes.

A.5 5' mRNA structure

ViennaRNA fold reports the minimum free energy of the 5' window of length w :

$$\text{MFE}_g = \min_{s \in \mathcal{S}} E(s, \text{seq}_g[1..w]), \quad (25)$$

over the secondary-structure space $\mathcal{S}$; less negative MFE near the start codon is associated with higher initiation efficiency.

A.6 Bootstrap confidence intervals

Paired bootstrap over $B = 10,000$ resamples of the gene axis gives the percentile interval

$$\text{CI}_{95} = [Q_{0.025}(\Delta_b), Q_{0.975}(\Delta_b)], \quad \Delta_b = \bar{z}_b^{\text{CodonOpt}} - \bar{z}_b^{\text{ICOR}}. \quad (26)$$

B Per-gene benchmark table (ICOR 40-gene panel)

All six methods, all 40 genes. `pred_z` is the predicted-expression z-score under our CNN; `original` = wild type. Full machine-readable record: `results/icor_headtohead_per_gene.json`.

Table 10: Per-gene codon-optimization benchmark, all methods. Cells show CAI/predicted-expression z .

Gene	original CAI/ z	icor CAI/ z	gensmart CAI/ z	HFC CAI/ z	BFC CAI/ z	URC CAI/ z	ERC CAI/ z
BIRC5	0.597/-0.19	0.863/0.67	0.731/0.14	0.997/0.52	0.630/-0.09	0.538/-0.56	0.674/0.07
BRAF1	0.528/-0.67	0.841/0.15	0.738/0.40	0.998/0.68	0.653/-0.08	0.546/-0.48	0.592/-0.19
CAV1	0.661/0.20	0.892/0.12	0.765/0.42	0.999/0.21	0.662/0.04	0.561/-0.51	0.686/0.01
CD80	0.554/-0.36	0.875/0.78	0.742/0.30	0.998/0.57	0.650/0.07	0.532/-0.51	0.620/-0.11
CDK1	0.523/-0.43	0.891/0.78	0.731/0.37	0.998/0.83	0.640/0.15	0.483/-0.24	0.607/-0.22
CEBPZ	0.573/-0.37	0.904/1.25	0.721/0.76	0.998/0.95	0.631/0.12	0.537/-0.53	0.600/0.33
CLN3	0.598/-0.48	0.847/-0.03	0.714/-0.90	0.998/-0.16	0.646/-0.68	0.510/-1.55	0.574/-1.18
CREB1	0.534/-0.02	0.862/1.42	0.751/1.14	0.998/1.37	0.666/0.46	0.559/0.21	0.629/0.26
CSNK1A1	0.543/-0.11	0.893/1.17	0.732/0.75	0.997/1.14	0.647/0.48	0.540/-0.54	0.631/-0.37
FGFR4	0.581/-0.27	0.863/0.83	0.719/-0.39	0.997/0.76	0.640/-0.70	0.511/-0.80	0.564/-0.95
GSK3B	0.530/-0.62	0.860/0.72	0.724/-0.30	0.997/0.37	0.675/0.14	0.537/-0.89	0.611/0.11
JUN	0.629/-0.55	0.851/0.60	0.738/0.36	0.998/1.03	0.631/0.71	0.532/-0.55	0.601/-0.06
KIF11	0.550/-0.12	0.873/1.50	0.721/1.10	0.998/1.43	0.638/0.41	0.549/0.55	0.585/0.42
LAMP1	0.584/-0.52	0.837/0.21	0.727/-0.70	0.998/0.08	0.653/-0.35	0.545/-0.50	0.607/-1.10
LEMD3	0.531/-0.28	0.852/1.50	0.728/0.88	0.997/0.93	0.651/0.37	0.524/-0.20	0.571/0.44
MAPK1	0.573/-0.10	0.892/0.58	0.744/0.77	0.998/0.77	0.663/0.26	0.504/-0.19	0.620/0.24
MAPKAPK5	0.550/0.20	0.906/1.05	0.740/0.50	0.997/1.12	0.664/0.66	0.552/-0.27	0.602/-0.24
NGFR	0.642/-0.20	0.861/0.61	0.742/0.26	0.998/0.55	0.667/-0.05	0.555/-0.78	0.619/0.44
NOC2L	0.590/0.08	0.866/0.85	0.717/-0.29	0.997/0.81	0.651/0.60	0.513/-0.65	0.574/-0.55
OPRM1	0.584/-0.92	0.862/-0.32	0.763/-0.50	0.998/-0.36	0.677/-0.65	0.544/-1.34	0.618/-0.93
PDCD11	0.561/0.19	0.869/1.01	0.710/0.39	0.998/0.84	0.629/0.02	0.537/-0.55	0.523/-0.50
PLK1	0.573/0.03	0.875/0.79	0.719/0.26	0.996/0.59	0.624/0.08	0.507/-0.51	0.573/-0.54
RPS6KB1	0.516/-0.74	0.889/1.10	0.720/0.23	0.997/0.88	0.632/0.37	0.542/-0.01	0.594/0.15
SMARCD1	0.562/-0.13	0.884/0.70	0.732/0.38	0.997/0.23	0.636/0.12	0.513/-0.66	0.602/-0.60
TAP1	0.570/-0.65	0.853/0.01	0.707/-0.43	0.998/0.10	0.642/-0.90	0.498/-1.16	0.555/-1.08
TAS2R10	0.517/-1.29	0.869/-0.50	0.731/-0.73	0.998/-0.72	0.686/-0.62	0.524/-1.12	0.624/-0.93
akt1	0.587/-0.17	0.895/0.92	—	0.997/0.68	—	—	—
emg1	0.531/-0.16	0.866/1.28	—	0.997/1.06	—	—	—
falvac-1	0.592/-0.15	0.888/0.80	—	0.999/0.94	—	—	—

continued

Gene	original CAI/z	icor CAI/z	gensmart CAI/z	HFC CAI/z	BFC CAI/z	URC CAI/z	ERC CAI/z
flt1	0.557/-1.01	0.864/-0.11	–	0.998/0.01	–	–	–
hpdf	0.664/-0.33	0.899/0.19	–	0.995/0.11	–	–	–
lck	0.630/-0.10	0.885/0.53	–	0.997/0.31	–	–	–
mmp13	0.665/0.05	0.848/0.54	–	0.997/0.50	–	–	–
npr1	0.612/-0.33	0.871/0.26	–	0.997/0.00	–	–	–
pa	0.517/-0.40	0.923/1.14	–	0.996/0.97	–	–	–
pak1	0.559/0.35	0.886/1.00	0.728/1.26	0.998/1.22	0.647/0.88	0.547/0.40	–
pea	0.743/-0.11	0.882/0.84	–	0.997/0.60	–	–	–
pim1	0.585/-0.39	0.890/0.69	–	0.997/0.65	–	–	–
ptp4a3	0.676/0.21	0.897/1.16	–	0.997/0.93	–	–	–
ubtf	0.577/0.09	0.893/0.57	–	0.997/0.61	–	–	–

C Seed-ensemble attribution audit (16 splits)

Per-seed held-out AUCs and the stability statistics reported in the body. Median held-out AUC = 0.749; split-half Spearman (8v8) = 0.187; median single-split pairwise Spearman = -0.056.

Seed	held-out AUC
101	0.778
102	0.883
103	0.811
104	0.760
105	0.709
106	0.694
107	0.709
108	0.811
109	0.619
110	0.625
111	0.613
112	0.769
113	0.820
114	0.745
115	0.736
116	0.754

Single-split pairwise Spearman values: 0.226, 0.116, 0.144, -0.081, -0.090, -0.089, -0.031, -0.138. Ensemble top-10 genes by mean permutation importance: TTN, MACF1, SYNE2, PEG3, CMYA5, TGFBR2, DCHS2, RYR2, RNF213, MUC16. The top genes are known long-gene passengers (TTN, MACF1, SYNE2 class); the rescue is rejected, as reported in the body.

D Replication contingency tables

D.1 paad_qcmg_uq_2016 ($n = 456$, CDKN2A-altered = 71)

Covariate	OR	p	a	table
KRAS	4.12	0.0003	65	[[65, 6], [279, 106]]
TP53_pathway	4.69	7.19e-07	60	[[60, 11], [207, 178]]
SWI_SNF	2.05	0.0621	13	[[13, 58], [38, 347]]
RTK_RAS_nonKRAS	0.28	0.3390	1	[[1, 70], [19, 366]]

D.2 pdac_msk_2024 ($n = 2336$, CDKN2A-altered = 546)

Covariate	OR	p	a	table
KRAS	1.91	0.0065	525	[[525, 21], [1663, 127]]
TP53_pathway	3.79	2.00e-20	500	[[500, 46], [1327, 463]]
SWI_SNF	0.76	0.0737	61	[[61, 485], [255, 1535]]
RTK_RAS_nonKRAS	1.21	0.4230	30	[[30, 516], [82, 1708]]

D.3 paad_qcmg_uq_2016 ($n = 456$, CDKN2A-altered = 71)

Covariate	OR	p	a	table
KRAS	4.12	0.0003	65	[[65, 6], [279, 106]]
TP53_pathway	4.69	7.19e-07	60	[[60, 11], [207, 178]]
SWI_SNF	2.05	0.0621	13	[[13, 58], [38, 347]]
RTK_RAS_nonKRAS	0.28	0.3390	1	[[1, 70], [19, 366]]

D.4 pdac_msk_2024 ($n = 2336$, CDKN2A-altered = 546)

Covariate	OR	p	a	table
KRAS	1.91	0.0065	525	[[525, 21], [1663, 127]]
TP53_pathway	3.79	2.00e-20	500	[[500, 46], [1327, 463]]
SWI_SNF	0.76	0.0737	61	[[61, 485], [255, 1535]]
RTK_RAS_nonKRAS	1.21	0.4230	30	[[30, 516], [82, 1708]]

D.5 paad_icgc ($n = 99$, CDKN2A-altered = 2)

Covariate	OR	p	a	table
KRAS	None	1.0000	2	[[2, 0], [92, 5]]
TP53_pathway	1.94	1.0000	1	[[1, 1], [33, 64]]
SWI_SNF	0.00	1.0000	0	[[0, 2], [8, 89]]
RTK_RAS_nonKRAS	0.00	1.0000	0	[[0, 2], [6, 91]]

D.6 paad_utsu_2015 ($n = 109$, **CDKN2A-altered** = 6)

Covariate	OR	p	a	table
KRAS	None	1.0000	6	[[6, 0], [94, 9]]
TP53_pathway	None	0.0280	6	[[6, 0], [51, 52]]
SWI_SNF	1.67	0.5120	1	[[1, 5], [11, 92]]
RTK_RAS_nonKRAS	2.93	0.2350	2	[[2, 4], [15, 88]]

D.7 paad_cptac_2021 ($n = 140$, **CDKN2A-altered** = 29)

Covariate	OR	p	a	table
KRAS	0.38	0.2760	27	[[27, 2], [108, 3]]
TP53_pathway	2.11	0.2240	25	[[25, 4], [83, 28]]
SWI_SNF	3.10	0.1270	5	[[5, 24], [7, 104]]
RTK_RAS_nonKRAS	1.29	1.0000	1	[[1, 28], [3, 108]]

D.8 paac_jhu_2014 ($n = 23$, **CDKN2A-altered** = 0)

Covariate	OR	p	a	table
KRAS	None	1.0000	0	[[0, 0], [0, 23]]
TP53_pathway	None	1.0000	0	[[0, 0], [4, 19]]
SWI_SNF	None	1.0000	0	[[0, 0], [3, 20]]
RTK_RAS_nonKRAS	None	1.0000	0	[[0, 0], [4, 19]]

D.9 pancreas_msk_2024 ($n = 395$, **CDKN2A-altered** = 93)

Covariate	OR	p	a	table
KRAS	11.29	0.0013	92	[[92, 1], [269, 33]]
TP53_pathway	3.21	0.0004	82	[[82, 11], [211, 91]]
SWI_SNF	0.87	0.8490	9	[[9, 84], [33, 269]]
RTK_RAS_nonKRAS	1.50	0.5460	5	[[5, 88], [11, 291]]

D.10 paad_msk_2025 ($n = 29$, **CDKN2A-altered** = 3)

Covariate	OR	p	a	table
KRAS	None	0.2680	3	[[3, 0], [15, 11]]
TP53_pathway	None	0.0996	3	[[3, 0], [11, 15]]
SWI_SNF	2.10	0.5150	1	[[1, 2], [5, 21]]
RTK_RAS_nonKRAS	0.00	1.0000	0	[[0, 3], [3, 23]]

D.11 paad_tcga ($n = 186$, CDKN2A-altered = 22)

Covariate	OR	p	a	table
KRAS	None	0.0013	22	[[22, 0], [114, 50]]
TP53_pathway	3.79	0.0200	18	[[18, 4], [89, 75]]
SWI_SNF	0.00	0.3700	0	[[0, 22], [13, 151]]
RTK_RAS_nonKRAS	3.08	0.1260	3	[[3, 19], [8, 156]]

E gnomAD r4 panel variants (top 150 by allele count)

Population-frequency audit of the detection panel. Full 9,410-row record: `results/gnomad_panel_variants.csv`.

Table 11: Top 150 panel variants by combined gnomAD allele count.

Gene	Variant	Consequence	rsID
CDKN2A	9-21968200-C-G	3_prime_UTR_variant	rs11515
TP53	17-7676325-CCCCCAGCCCTCCAGGT-C	intron_variant	rs59758982
TP53	17-7674797-T-C	intron_variant	rs1625895
TP53	17-7676154-G-C	missense_variant	rs1042522
TP53	17-7676483-G-C	5_prime_UTR_variant	rs1642785
KRAS	12-25209843-A-G	synonymous_variant	rs1137282
SMAD4	18-51058485-A-ATT	intron_variant	rs38638767
TP53	17-7674360-CT-C	intron_variant	rs75277459
CDKN2A	9-21968160-G-A	3_prime_UTR_variant	rs3088440
SMAD4	18-51058485-A-AT	intron_variant	rs38638767
TP53	17-7674109-G-A	intron_variant	rs12947788
SMAD4	18-51058485-AT-A	intron_variant	rs38638767
TP53	17-7676301-G-T	intron_variant	rs17883323
SMAD4	18-51059814-A-G	intron_variant	rs948589
SMAD4	18-51059974-C-T	intron_variant	rs948588
TP53	17-7667260-TA-T	intron_variant	rs34103303
TP53	17-7667260-T-TA	intron_variant	rs34103303
CDKN2A	9-21970917-C-T	missense_variant	rs3731249
TP53	17-7673183-G-A	intron_variant	rs77697176
KRAS	12-25209920-C-T	intron_variant	rs12313763
TP53	17-7674360-C-CT	intron_variant	rs75277459
SMAD4	18-51058485-A-ATTT	intron_variant	rs38638767
TP53	17-7674892-T-C	synonymous_variant	rs1800372
TP53	17-7673523-A-G	intron_variant	rs1800899
TP53	17-7676261-C-T	synonymous_variant	rs1800370
TP53	17-7674326-C-G	intron_variant	rs17880604
TP53	17-7670579-T-A	intron_variant	rs17880847
TP53	17-7666224-G-A	synonymous_variant	rs74351250
TP53	17-7669739-G-A	intron_variant	rs17881850
CDKN2A	9-21968713-C-A	3_prime_UTR_variant	rs34011899
SMAD4	18-51065392-G-A	intron_variant	rs11875522
TP53	17-7674360-CTT-C	intron_variant	rs75277459
TP53	17-7674828-T-C	intron_variant	rs34949160
CDKN2A	9-21968653-G-C	3_prime_UTR_variant	rs11153278
SMAD4	18-51078478-C-T	3_prime_UTR_variant	rs11663402
TP53	17-7667259-T-A	intron_variant	rs73979529
SMAD4	18-51049378-T-G	intron_variant	rs19300074
KRAS	12-25227160-A-C	intron_variant	rs11590833
SMAD4	18-51048790-G-A	synonymous_variant	rs14598861

Gene	Variant	Consequence	rsID
TP53	17-7676323-GCCCCCAGCCC-G	intron_variant	rs77903581
TP53	17-7676336-CCAGGT-C	intron_variant	rs77663963
TP53	17-7669763-C-G	intron_variant	rs17883043
KRAS	12-25228889-T-A	intron_variant	rs14785017
CDKN2A	9-21974641-C-G	intron_variant	rs45456595
SMAD4	18-51058485-ATT-A	intron_variant	rs38638767
SMAD4	18-51047319-A-G	intron_variant	rs77389132
KRAS	12-25245228-A-C	intron_variant	rs19971681
CDKN2A	9-21974852-G-A	5_prime_UTR_variant	rs14448158
SMAD4	18-51065650-T-C	intron_variant	rs28539779
KRAS	12-25230575-C-T	intron_variant	rs12228878
CDKN2A	9-21969810-G-C	intron_variant	rs14223148
CDKN2A	9-21974463-AC-A	intron_variant	rs20012949
CDKN2A	9-21974459-CTGT-C	intron_variant	rs20182019
CDKN2A	9-21974451-CGAGAGA-C	intron_variant	rs20050482
KRAS	12-25228915-C-T	intron_variant	rs34088434
SMAD4	18-51076571-G-C	intron_variant	rs79051374
SMAD4	18-51076877-G-GT	3_prime_UTR_variant	rs10471733
TP53	17-7673394-C-T	intron_variant	rs37159226
TP53	17-7676278-G-A	splice_region_variant	rs35117667
SMAD4	18-51065553-T-C	synonymous_variant	rs1801250
CDKN2A	9-21974860-C-G	5_prime_UTR_variant	rs53159773
KRAS	12-25228916-G-A	intron_variant	rs14829046
SMAD4	18-51058050-G-A	intron_variant	rs11463401
SMAD4	18-51067219-CTT-C	intron_variant	rs20168328
SMAD4	18-51054738-A-T	intron_variant	rs14954101
SMAD4	18-51078381-A-G	missense_variant	rs14975532
KRAS	12-25209828-TTTC-T	inframe_deletion	rs39751704
TP53	17-7674360-CTTT-C	intron_variant	rs75277459
CDKN2A	9-21970980-C-A	missense_variant	rs6413464
SMAD4	18-51058485-A-ATTTT	intron_variant	rs38638767
CDKN2A	9-21971212-C-G	splice_region_variant	rs52938097
CDKN2A	9-21968733-G-A	missense_variant	rs18104451
TP53	17-7676230-G-A	missense_variant	rs1800371
KRAS	12-25228885-C-T	intron_variant	rs14456242
SMAD4	18-51058244-AGTATGTACATACTTTAAAAAATCTTTTAAATAGTTGAGAAAAAAGTA	splice_acceptor_variant	rs15991896
SMAD4	18-51067247-T-C	intron_variant	rs13974167
SMAD4	18-51054720-A-AT	intron_variant	rs96177359
TP53	17-7666228-C-G	missense_variant	rs14436692
TP53	17-7674811-C-T	intron_variant	rs17884607
CDKN2A	9-21970819-G-A	3_prime_UTR_variant	rs56068638
CDKN2A	9-21974829-C-T	5_prime_UTR_variant	rs19139414
CDKN2A	9-21974462-T-C	intron_variant	rs14760278
CDKN2A	9-21971185-T-G	missense_variant	rs20120889
CDKN2A	9-21974525-A-T	intron_variant	rs15044751
CDKN2A	9-21970929-G-A	missense_variant	rs11615089
CDKN2A	9-21974439-C-A	intron_variant	rs14514390
KRAS	12-25245361-T-C	synonymous_variant	rs14740641
SMAD4	18-51054775-A-G	splice_region_variant	rs18117886
TP53	17-7662034-C-A	intron_variant	rs14027275
SMAD4	18-51055047-T-C	intron_variant	rs37119356
SMAD4	18-51067009-T-C	intron_variant	rs18633216
CDKN2A	9-21994102-C-G	intron_variant	rs11651282
CDKN2A	9-21970986-C-G	missense_variant	rs14617913
KRAS	12-25228911-T-C	intron_variant	rs77460055
CDKN2A	9-21970827-T-C	3_prime_UTR_variant	rs78147791
SMAD4	18-51078207-G-C	intron_variant	rs37531366

Gene	Variant	Consequence	rsID
CDKN2A	9-21969808-C-CG	intron_variant	rs10140576
TP53	17-7674360-C-CTT	intron_variant	rs75277459
CDKN2A	9-21971041-C-T	missense_variant	rs19988800
TP53	17-7674319-G-C	intron_variant	–
CDKN2A	9-21974841-G-A	5_prime_UTR_variant	rs76424471
SMAD4	18-51078223-T-A	intron_variant	rs36812112
SMAD4	18-51049347-C-T	intron_variant	rs20077869
CDKN2A	9-21968304-C-T	intron_variant	rs37659110
TP53	17-7674164-G-A	intron_variant	rs17880172
CDKN2A	9-21974526-G-A	intron_variant	rs75256133
KRAS	12-25228928-G-A	intron_variant	rs14051030
SMAD4	18-51076583-A-G	intron_variant	rs57797067
SMAD4	18-51066962-AAAATT-A	intron_variant	rs56325334
KRAS	12-25215528-T-C	synonymous_variant	rs4362222
SMAD4	18-51058099-A-G	intron_variant	rs37585699
SMAD4	18-51058485-ATTT-A	intron_variant	rs38638767
SMAD4	18-51054891-C-T	missense_variant	rs14074323
KRAS	12-25215476-C-T	missense_variant	rs20097034
SMAD4	18-51076721-C-T	synonymous_variant	rs14048710
TP53	17-7674259-T-C	missense_variant	rs14434071
TP53	17-7675029-G-A	intron_variant	rs56181208
TP53	17-7676532-G-A	synonymous_variant	rs1800369
KRAS	12-25215446-T-G	missense_variant	rs20117065
TP53	17-7670695-G-A	synonymous_variant	rs15029382
CDKN2A	9-21970990-A-T	missense_variant	rs6413463
TP53	17-7673751-C-T	missense_variant	rs55819519
KRAS	12-25215471-A-T	stop_gained	rs37316952
TP53	17-7676589-C-T	synonymous_variant	rs14345827
SMAD4	18-51078479-G-A	3_prime_UTR_variant	rs14868703
CDKN2A	9-21994263-G-A	synonymous_variant	rs37436079
CDKN2A	9-21970954-C-T	synonymous_variant	rs75158639
CDKN2A	9-21971162-T-C	missense_variant	rs75675025
SMAD4	18-51048865-G-A	intron_variant	rs20077260
TP53	17-7670549-A-G	intron_variant	rs37590515
TP53	17-7674360-CTTTT-C	intron_variant	rs75277459
KRAS	12-25209929-T-C	intron_variant	rs20018910
SMAD4	18-51049362-G-C	intron_variant	rs20192471
CDKN2A	9-21968223-G-C	missense_variant	rs37562841
KRAS	12-25227260-T-C	synonymous_variant	rs37092066
TP53	17-7676387-C-T	missense_variant	rs20175335
TP53	17-7670630-C-G	missense_variant	rs35993958
CDKN2A	9-21970975-C-T	splice_donor_variant	rs19990189
TP53	17-7673312-A-C	missense_variant	rs3021068
TP53	17-7676507-A-G	5_prime_UTR_variant	rs18474315
TP53	17-7674327-G-A	intron_variant	rs37490773
CDKN2A	9-21968696-G-A	3_prime_UTR_variant	rs54282670
TP53	17-7670732-G-A	intron_variant	rs36869191
CDKN2A	9-21968728-C-T	synonymous_variant	rs18912716
SMAD4	18-51055002-T-C	intron_variant	rs77652320
TP53	17-7673870-A-G	intron_variant	rs11330258
TP53	17-7674169-G-A	intron_variant	rs17881780
TP53	17-7676281-G-A	intron_variant	rs20221726
CDKN2A	9-21974858-C-T	5_prime_UTR_variant	rs73088167
KRAS	12-25245295-G-A	synonymous_variant	rs11362314

F CIViC evidence rows for the driver panel

Curated clinical evidence supporting panel genes. Full 1,153-row record: [results/civic_panel_evidence_rows.csv](#). Pancreatic-relevant rows first. Of 1153 evidence rows, 65 are pancreatic-cancer relevant.

Table 12: CIViC evidence rows (pancreatic-relevant first, top 150 shown).

EID	Gene	Level	Type	Significance	Panc.	Molecular profile
EID622	KRAS	B	PREDICTIVE	RESISTANCE	1	KRAS G12/G13
EID1219	KRAS	B	PREDICTIVE	RESISTANCE	1	KRAS Exon 2 Mutation
EID1298	KRAS	B	PROGNOSTIC	POOR_OUTCOME	1	KRAS G12R
EID1300	KRAS	B	PROGNOSTIC	POOR_OUTCOME	1	KRAS G12D
EID916	KRAS	B	PREDICTIVE	RESISTANCE	1	KRAS Exon 2 Mutation
EID917	KRAS	B	PROGNOSTIC	POOR_OUTCOME	1	KRAS Exon 2 Mutation
EID990	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12/G13
EID1311	KRAS	B	PROGNOSTIC	POOR_OUTCOME	1	KRAS G12D
EID1732	KRAS	B	PROGNOSTIC	POOR_OUTCOME	1	KRAS G12D
EID7180	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS Activating Mutation
EID11680	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12C
EID12938	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D
EID13294	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D
EID7492	KRAS	C	PREDICTIVE	SENSITIVITYRESPONS	1	ATP1B1::NRG1 Fusion AND KRAS Wildtype
EID7493	KRAS	C	PREDICTIVE	SENSITIVITYRESPONS	1	APP::NRG1 Fusion AND KRAS Wildtype
EID11710	KRAS	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND KRAS G12R AND CDKN2A Loss
EID11711	KRAS	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND KRAS G12R AND CDKN2A Loss
EID11713	KRAS	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND CDKN2A p16 R80*
EID11714	KRAS	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND GNAS R201C AND NF1 D1976fs
EID937	KRAS	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D
EID3959	KRAS	D	PREDICTIVE	RESISTANCE	1	KRAS G12D
EID1003	KRAS	D	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS Mutation
EID7171	KRAS	D	PROGNOSTIC	POOR_OUTCOME	1	KRAS G12D
EID9602	KRAS	D	PREDICTIVE	RESISTANCE	1	KRAS KRAS4A underexpression
EID12136	KRAS	D	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D
EID11548	KRAS	D	ONCOGENIC	ONCOGENICITY	1	KRAS G12D AND ERBB2 S310F
EID11550	KRAS	D	ONCOGENIC	ONCOGENICITY	1	KRAS G12D AND ERBB2 S423R
EID11552	KRAS	D	ONCOGENIC	ONCOGENICITY	1	ERBB2 R678Q AND KRAS G12D
EID11554	KRAS	D	ONCOGENIC	ONCOGENICITY	1	KRAS G12D AND ERBB2 Q679L
EID11556	KRAS	D	ONCOGENIC	ONCOGENICITY	1	KRAS G12D AND ERBB2 E717D
EID11558	KRAS	D	ONCOGENIC	ONCOGENICITY	1	ERBB2 L755S AND KRAS G12D
EID11560	KRAS	D	ONCOGENIC	ONCOGENICITY	1	ERBB2 V777L AND KRAS G12D
EID11562	KRAS	D	ONCOGENIC	ONCOGENICITY	1	ERBB2 V842I AND KRAS G12D
EID11572	KRAS	D	PREDICTIVE	SENSITIVITYRESPONS	1	ERBB2 V777L AND KRAS G12C

continued

EID	Gene	Level	Type	Significance	Panc.	Molecular profile
EID11573	KRAS	D	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12C AND ERBB2 V842I
EID7181	KRAS	D	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D
EID7182	KRAS	D	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS Activating Mutation
EID7183	KRAS	D	PREDICTIVE	RESISTANCE	1	KRAS G12
EID8041	KRAS	D	PREDISPOSING	NA	1	KRAS V14I
EID7184	KRAS	D	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS Activating Mutation
EID8282	TP53	B	PREDICTIVE	SENSITIVITYRESPONS	1	TP53 Mutation
EID11710	TP53	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND KRAS G12R AND CDKN2A Loss
EID11711	TP53	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND KRAS G12R AND CDKN2A Loss
EID11712	TP53	C	PREDICTIVE	SENSITIVITYRESPONS	1	BRAF V600E AND ERBB2 Amplification AND S
EID7299	CDKN2A	B	PREDICTIVE	SENSITIVITYRESPONS	1	CDKN2A Mutation
EID11653	CDKN2A	B	PREDICTIVE	SENSITIVITYRESPONS	1	CDKN2A Loss OR CDKN2A Mutation
EID11710	CDKN2A	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND KRAS G12R AND CDKN2A Loss
EID11711	CDKN2A	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND KRAS G12R AND CDKN2A Loss
EID11713	CDKN2A	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND CDKN2A p16 R80*
EID6440	SMAD4	B	PROGNOSTIC	BETTER_OUTCOME	1	SMAD4 Underexpression
EID6445	SMAD4	B	PROGNOSTIC	POOR_OUTCOME	1	SMAD4 Underexpression
EID511	SMAD4	B	PROGNOSTIC	POOR_OUTCOME	1	SMAD4 Mutation
EID1306	SMAD4	B	PROGNOSTIC	BETTER_OUTCOME	1	SMAD4 Expression
EID1412	SMAD4	B	PROGNOSTIC	POOR_OUTCOME	1	SMAD4 LOSS
EID7179	SMAD4	C	PROGNOSTIC	NA	1	SMAD4 Expression
EID11711	SMAD4	C	PREDICTIVE	SENSITIVITYRESPONS	1	KRAS G12D AND KRAS G12R AND CDKN2A Loss
EID11712	SMAD4	C	PREDICTIVE	SENSITIVITYRESPONS	1	BRAF V600E AND ERBB2 Amplification AND S
EID7384	BRCA1	A	PREDICTIVE	SENSITIVITYRESPONS	1	BRCA1 Loss-of-function
EID5915	BRCA1	B	PREDICTIVE	SENSITIVITYRESPONS	1	BRCA1 Mutation
EID5914	BRCA1	B	PREDICTIVE	SENSITIVITYRESPONS	1	BRCA1 Mutation
EID5932	BRCA1	B	PREDICTIVE	SENSITIVITYRESPONS	1	BRCA1 Mutation
EID5934	BRCA1	B	PREDICTIVE	SENSITIVITYRESPONS	1	BRCA1 Mutation
EID11317	BRCA1	B	PREDICTIVE	SENSITIVITYRESPONS	1	BRCA1 Mutation OR BRCA2 Mutation OR PALB
EID9291	BRCA1	B	PREDICTIVE	SENSITIVITYRESPONS	1	BRCA1 Mutation
EID7435	BRCA1	C	PREDICTIVE	SENSITIVITYRESPONS	1	BRCA1 Mutation
EID993	KRAS	A	PREDICTIVE	RESISTANCE	0	KRAS Exon 2 Mutation
EID2998	KRAS	A	PREDICTIVE	RESISTANCE	0	KRAS Mutation
EID5345	KRAS	A	PREDICTIVE	RESISTANCE	0	KRAS Mutation
EID9431	KRAS	A	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G12C
EID11262	KRAS	A	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Wildtype
EID11264	KRAS	A	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Wildtype
EID12063	KRAS	A	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G12C
EID12264	KRAS	A	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G12C
EID12818	KRAS	A	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID2010	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12A
EID2011	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12A
EID2241	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID2243	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID2244	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID2245	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D

continued

EID	Gene	Level	Type	Significance	Panc.	Molecular profile
EID2248	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID2249	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID2259	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12C
EID2260	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12C
EID2275	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12S
EID2276	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12S
EID2806	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS Q61P
EID2808	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS Q61E
EID3879	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G13D
EID3882	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G13D
EID3884	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G13D
EID3916	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12V
EID3919	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12V
EID3938	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID3951	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID3952	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID3953	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID3954	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12D
EID4005	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12S
EID4508	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G13V
EID8068	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G12C
EID306	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G13D
EID336	KRAS	B	PROGNOSTIC	BETTER_OUTCOME	0	KRAS G12
EID487	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS G12
EID488	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS G13
EID489	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS Q61
EID999	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID1139	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS Mutation
EID1181	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G13D
EID1700	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS Mutation
EID3704	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS Mutation
EID3917	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS Q61
EID4867	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID4868	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID4996	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID4997	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID6974	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G13D
EID8031	KRAS	B	DIAGNOSTIC	POSITIVE	0	KRAS G12D
EID8301	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID9598	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS KRAS4A underexpres- sion
EID9599	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS KRAS4A underexpres- sion
EID9600	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS KRAS4A underexpres- sion
EID11055	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G12 OR KRAS G13 OR KRAS Q61
EID11061	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	NRAS Mutation OR KRAS Mu- tation OR BRAF M
EID11328	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	NRAS Mutation OR KRAS Mu- tation OR RAF1 M
EID11329	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	NRAS Mutation OR KRAS Mu- tation OR RAF1 M
EID11652	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G12C
EID13009	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS G12C
EID13126	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID8	KRAS	B	DIAGNOSTIC	POSITIVE	0	KRAS G12

continued

EID	Gene	Level	Type	Significance	Panc.	Molecular profile
EID35	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12/G13
EID134	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12/G13
EID215	KRAS	B	DIAGNOSTIC	POSITIVE	0	KRAS G12
EID228	KRAS	B	DIAGNOSTIC	POSITIVE	0	KRAS G12D
EID250	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS Exon 2 Mutation
EID251	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12A
EID337	KRAS	B	PROGNOSTIC	NA	0	KRAS G12
EID377	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS G12
EID378	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS G12
EID664	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS RS61764370
EID872	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS RS61764370
EID987	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS G12V
EID997	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS Mutation
EID1180	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G13D
EID1208	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS Exon 2 Mutation
EID1215	KRAS	B	PROGNOSTIC	BETTER_OUTCOME	0	KRAS G12D
EID1216	KRAS	B	PREDICTIVE	RESISTANCE	0	KRAS G12C
EID1217	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS G12C
EID1218	KRAS	B	PREDICTIVE	SENSITIVITYRESPONS	0	KRAS Mutation
EID1596	KRAS	B	PROGNOSTIC	POOR_OUTCOME	0	KRAS Mutation

G DepMap 24Q4 panel gene-effect sample

Chronos gene-effect scores for the seven panel genes across screened lines (sample of 150 of 1,178). Full record: `results/depmap_panel_subset.csv`.

Table 13: DepMap Chronos scores, sample.

Model	Lineage	CDKN2A	KRAS	PCNA	PSMA1	RPS3	SMAD4	TP53
ACH-000001	Ovary/Fallopian Tu	0.00	-0.31	-2.95	-2.34	-2.45	0.05	-0.00
ACH-000004	Myeloid	0.19	-0.24	-3.27	-2.60	-2.02	-0.16	-0.07
ACH-000005	Myeloid	0.05	-0.71	-2.74	-2.87	-2.17	-0.18	-0.14
ACH-000007	Bowel	0.46	-2.19	-2.29	-1.79	-3.24	-0.48	1.39
ACH-000009	Bowel	0.14	-1.31	-3.52	-2.30	-2.44	-0.31	0.08
ACH-000011	Bladder/Urinary Tr	0.30	-0.03	-2.94	-1.96	-2.47	-0.07	1.28
ACH-000012	Lung	0.05	-0.15	-3.39	-2.17	-1.99	-0.12	0.20
ACH-000013	Ovary/Fallopian Tu	-0.04	-0.15	-3.15	-2.31	-2.67	0.11	0.04
ACH-000014	Skin	0.43	-0.26	-2.00	-2.44	-2.12	-0.13	1.51
ACH-000015	Lung	0.16	-0.27	-2.86	-2.34	-2.40	-0.14	0.33
ACH-000017	Breast	0.10	-0.49	-2.62	-2.05	-2.56	0.21	-0.04
ACH-000018	Bladder/Urinary Tr	0.17	-0.65	-2.68	-2.50	-2.12	-0.15	0.06
ACH-000019	Breast	0.24	-0.52	-3.07	-2.87	-2.76	0.02	0.46
ACH-000021	Lung	0.23	-0.43	-2.66	-2.28	-2.48	-0.08	0.24
ACH-000022	Pancreas	0.06	-2.03	-2.85	-1.95	-2.13	-0.09	0.32
ACH-000023	Pancreas	0.00	-1.62	-2.95	-1.66	-2.00	0.18	0.16
ACH-000024	Lymphoid	0.01	-0.93	-3.11	-2.39	-2.49	0.18	0.14
ACH-000025	CNS/Brain	0.20	-0.40	-3.53	-2.08	-2.36	-0.22	0.08
ACH-000026	Bladder/Urinary Tr	0.22	-0.40	-2.85	-2.27	-2.41	-0.06	0.96
ACH-000028	Breast	0.18	-0.22	-2.16	-1.97	-2.23	0.12	0.75
ACH-000029	Lung	0.12	-0.17	-2.63	-2.28	-2.52	-0.12	0.21
ACH-000030	Lung	0.10	-0.09	-2.98	-1.91	-2.58	-0.06	-0.31
ACH-000035	Lung	0.43	-0.44	-3.27	-1.82	-3.34	-0.15	0.09
ACH-000036	CNS/Brain	0.20	-0.37	-2.36	-2.12	-2.55	0.19	0.75
ACH-000037	Soft Tissue	0.08	-0.32	-0.91	-2.92	-3.44	0.13	0.26

continued

Model	Lineage	CDKN2A	KRAS	PCNA	PSMA1	RPS3	SMAD4	TP53
ACH-000039	Bone	-0.10	-0.71	-2.64	-2.21	-2.51	0.02	0.08
ACH-000040	CNS/Brain	-0.12	-0.29	-2.50	-2.53	-2.48	-0.07	0.65
ACH-000041	Bone	0.17	-0.53	-2.98	-2.45	-2.41	-0.15	0.15
ACH-000042	Pancreas	0.08	-2.84	-2.23	-2.21	-2.52	-0.07	-0.16
ACH-000045	Myeloid	0.53	-0.36	-1.52	-2.07	-2.35	-0.08	0.80
ACH-000047	Esophagus/Stomach	0.21	-0.11	-3.52	-2.09	-2.39	-0.05	0.14
ACH-000048	Ovary/Fallopian Tu	0.13	-0.29	-1.99	-2.41	-2.32	-0.17	0.26
ACH-000052	Bone	0.12	-0.57	-2.43	-2.40	-2.58	-0.24	-0.01
ACH-000053	Lymphoid	-0.07	-0.31	-3.71	-2.96	-2.50	-0.04	0.43
ACH-000054	Soft Tissue	0.31	-0.29	-2.15	-2.43	-2.55	-0.18	1.36
ACH-000055	CNS/Brain	-0.11	-0.17	-2.55	-2.15	-2.37	-0.64	0.36
ACH-000060	Pancreas	-0.04	-2.02	-2.80	-2.65	-2.45	-0.10	0.31
ACH-000067	CNS/Brain	0.04	-0.11	-3.38	-2.60	-3.08	0.01	0.30
ACH-000070	Lymphoid	0.20	-0.38	-2.60	-2.35	-2.32	-0.05	0.76
ACH-000074	Myeloid	-0.39	-0.43	-2.95	-2.80	-2.47	-0.19	0.42
ACH-000075	CNS/Brain	0.13	-0.44	-1.91	-2.57	-2.75	-0.15	0.65
ACH-000076	Myeloid	0.60	-0.60	-2.77	-2.41	-2.48	-0.16	0.04
ACH-000077	Lymphoid	0.43	-0.43	-2.74	-2.16	-3.20	0.03	0.44
ACH-000078	Peripheral Nervous	0.02	-0.45	-2.27	-2.02	-2.47	-0.26	0.77
ACH-000082	Bone	-0.07	-0.29	-3.33	-2.68	-2.35	-0.36	0.19
ACH-000085	Pancreas	0.01	-0.79	-2.36	-2.31	-2.31	-0.10	0.22
ACH-000086	Pleura	0.19	-0.23	-2.14	-2.67	-2.47	-0.18	0.38
ACH-000087	Bone	-0.02	-0.45	-2.86	-2.46	-2.06	-0.30	0.18
ACH-000090	Prostate	-0.09	-0.50	-3.00	-2.45	-2.48	-0.06	0.02
ACH-000092	Pleura	0.21	-0.30	-3.23	-2.92	-2.51	-0.13	0.13
ACH-000093	Pancreas	0.42	-1.35	-2.46	-2.21	-2.37	0.40	2.09
ACH-000094	Pancreas	-0.29	-2.63	-2.41	-2.76	-2.52	-0.25	0.02
ACH-000095	CNS/Brain	-0.12	-0.21	-2.27	-2.07	-2.51	-0.44	0.62
ACH-000096	Kidney	-0.02	-0.11	-2.62	-2.31	-2.27	-0.24	1.87
ACH-000097	Breast	0.33	-0.91	-2.95	-2.08	-2.69	0.15	1.11
ACH-000098	CNS/Brain	0.09	-0.24	-2.74	-2.35	-2.84	-0.19	0.25
ACH-000099	Peripheral Nervous	0.08	-0.71	-2.98	-2.27	-2.69	-0.25	0.54
ACH-000100	Soft Tissue	0.17	-0.39	-2.75	-2.35	-2.20	-0.31	0.07
ACH-000101	Lymphoid	0.18	-0.06	-2.06	-1.73	-1.94	0.05	0.16
ACH-000103	Ovary/Fallopian Tu	0.07	-0.31	-2.61	-2.65	-2.77	-0.50	0.05
ACH-000106	Lymphoid	0.41	-0.46	-2.81	-2.40	-2.32	-0.31	0.50
ACH-000108	Pancreas	0.04	-1.12	-3.18	-2.40	-2.57	-0.02	-0.08
ACH-000111	Breast	0.30	-1.11	-2.07	-1.62	-3.45	-0.06	0.06
ACH-000113	Myeloid	0.46	-0.05	-2.51	-1.59	-2.40	-0.09	0.89
ACH-000114	Pancreas	0.27	-2.55	-3.09	-2.12	-2.27	0.01	0.20
ACH-000115	Prostate	-0.28	-0.43	-2.87	-2.33	-2.20	-0.32	0.02
ACH-000116	Ovary/Fallopian Tu	-0.07	-0.40	-3.00	-2.23	-2.63	0.19	0.11
ACH-000118	Pancreas	0.23	-0.39	-3.42	-2.07	-2.65	-0.20	0.23
ACH-000120	Peripheral Nervous	0.08	-0.38	-2.23	-1.76	-2.06	-0.47	0.75
ACH-000123	Ovary/Fallopian Tu	0.45	-0.24	-2.85	-2.29	-2.28	-0.21	1.08
ACH-000124	Lymphoid	0.43	-0.77	-2.36	-2.21	-3.01	-0.25	1.03
ACH-000127	Bladder/Urinary Tr	0.12	-0.23	-2.34	-1.92	-2.37	-0.24	0.30
ACH-000128	CNS/Brain	0.10	-0.27	-2.40	-2.53	-2.80	-0.22	-0.03
ACH-000132	Ovary/Fallopian Tu	0.07	-0.56	-2.72	-2.23	-2.73	0.09	0.25
ACH-000133	Fibroblast	0.20	-0.76	-2.59	-2.28	-2.50	-0.07	0.10
ACH-000137	CNS/Brain	0.20	-0.31	-2.39	-2.46	-2.71	-0.17	0.03
ACH-000138	Pancreas	0.18	-1.83	-2.97	-2.22	-2.52	0.07	0.17
ACH-000139	Pancreas	0.18	-0.61	-2.85	-1.69	-2.09	-0.20	-0.15
ACH-000141	Biliary Tract	0.18	-0.23	-2.77	-2.34	-2.59	-0.42	0.07
ACH-000142	Bladder/Urinary Tr	0.33	-0.39	-3.26	-1.89	-2.26	0.39	0.27
ACH-000143	Lung	-0.10	-0.53	-3.06	-2.66	-2.02	-0.17	0.30
ACH-000144	Esophagus/Stomach	-0.07	-0.10	-3.14	-2.72	-2.39	0.09	0.16

continued

Model	Lineage	CDKN2A	KRAS	PCNA	PSMA1	RPS3	SMAD4	TP53
ACH-000145	Soft Tissue	0.17	-0.41	-2.91	-2.46	-2.53	-0.10	0.17
ACH-000146	Myeloid	-0.05	-0.40	-3.23	-2.39	-2.27	-0.08	0.16
ACH-000147	Breast	0.09	-0.32	-3.06	-1.68	-2.64	-0.10	0.22
ACH-000148	Breast	0.21	-0.40	-3.14	-2.19	-2.41	0.28	0.42
ACH-000149	Peripheral Nervous	0.29	-0.35	-2.60	-3.31	-3.50	0.06	1.37
ACH-000151	Lymphoid	0.10	-0.49	-2.89	-2.44	-2.12	0.25	0.30
ACH-000152	CNS/Brain	0.28	-0.25	-2.02	-2.68	-2.64	0.20	0.11
ACH-000153	Pleura	0.14	-0.77	-2.77	-1.84	-2.63	-0.04	1.29
ACH-000155	Pancreas	0.19	-2.25	-3.19	-1.73	-2.40	0.04	0.16
ACH-000159	Kidney	0.31	-0.12	-2.79	-2.74	-2.92	0.04	1.07
ACH-000160	CNS/Brain	0.22	-0.30	-1.54	-1.89	-2.24	-0.18	1.65
ACH-000161	Lung	0.63	-0.22	-2.83	-2.30	-2.82	0.09	0.95
ACH-000163	Head and Neck	0.05	-1.64	-3.25	-2.08	-2.27	-0.13	0.06
ACH-000164	Pancreas	0.26	-1.66	-2.82	-2.15	-2.45	-0.26	0.26
ACH-000167	Lymphoid	0.17	-0.29	-2.53	-2.58	-1.94	-0.12	0.36
ACH-000168	Myeloid	0.34	-2.29	-3.36	-2.38	-2.49	-0.17	0.10
ACH-000169	Soft Tissue	-0.09	-0.30	-2.60	-2.27	-2.19	-0.28	0.33
ACH-000172	Soft Tissue	-0.17	-0.51	-2.19	-2.42	-2.55	0.19	1.57
ACH-000174	Thyroid	0.20	-1.91	-2.52	-2.43	-2.75	-0.38	0.29
ACH-000176	Lung	-0.17	-0.26	-2.49	-2.89	-2.35	-0.08	0.18
ACH-000178	Pancreas	0.35	-1.16	-2.68	-2.09	-2.47	0.09	0.33
ACH-000181	Head and Neck	0.24	-0.36	-3.32	-2.21	-2.84	0.09	0.19
ACH-000182	Ampulla of Vater	0.13	-1.76	-2.81	-2.60	-2.42	-0.07	0.13
ACH-000183	Lymphoid	0.16	-0.21	-3.06	-1.97	-2.17	0.04	0.24
ACH-000187	Lung	0.11	-0.52	-2.34	-2.67	-2.33	-0.06	-0.03
ACH-000188	Head and Neck	0.14	-0.04	-2.95	-2.64	-2.07	-0.02	0.25
ACH-000189	Kidney	0.06	-0.40	-2.74	-2.28	-2.57	-0.12	1.43
ACH-000190	Myeloid	0.30	-0.28	-3.07	-2.17	-2.73	-0.21	0.10
ACH-000191	Thyroid	0.32	-0.15	-1.79	-1.86	-2.86	-0.12	0.19
ACH-000192	Uterus	0.14	-0.18	-2.76	-3.23	-3.06	0.20	0.32
ACH-000195	Myeloid	0.09	-0.42	-2.60	-3.09	-2.67	0.04	-0.20
ACH-000198	Myeloid	0.12	-0.47	-2.15	-2.20	-2.49	-0.06	0.67
ACH-000200	CNS/Brain	0.16	-0.30	-2.42	-2.47	-2.72	-0.37	2.32
ACH-000202	Bowel	0.27	-0.55	-2.76	-2.70	-1.85	-0.18	-0.14
ACH-000203	Peripheral Nervous	0.14	-1.03	-2.60	-2.45	-2.44	-0.05	0.42
ACH-000204	Lymphoid	0.02	-0.26	-3.31	-2.55	-2.77	-0.14	0.28
ACH-000206	Lymphoid	0.16	-0.31	-2.56	-3.01	-2.81	-0.29	-0.01
ACH-000207	Head and Neck	0.08	-0.44	-3.53	-2.15	-2.30	0.13	0.10
ACH-000208	CNS/Brain	-0.07	-0.30	-2.70	-2.56	-2.16	0.10	0.60
ACH-000209	Biliary Tract	0.28	-0.52	-2.75	-2.41	-3.17	0.15	1.73
ACH-000210	Bone	0.21	-0.83	-2.53	-2.49	-2.47	-0.01	0.47
ACH-000211	CNS/Brain	0.16	-0.27	-3.28	-2.06	-2.33	-0.19	0.23
ACH-000212	Breast	0.17	-0.25	-3.18	-2.14	-2.88	-0.41	0.18
ACH-000213	Pancreas	0.37	-1.97	-3.23	-2.64	-2.34	0.34	0.36
ACH-000215	CNS/Brain	0.38	-0.15	-2.80	-2.22	-3.07	0.12	0.59
ACH-000216	Esophagus/Stomach	0.16	-0.19	-3.18	-2.41	-2.57	-0.11	0.32
ACH-000217	Liver	0.14	-0.36	-3.05	-2.26	-2.17	-0.02	0.14
ACH-000219	Skin	0.26	-0.26	-2.74	-1.88	-3.02	-0.19	0.66
ACH-000221	Liver	0.20	-0.78	-1.65	-1.97	-2.13	0.04	0.07
ACH-000222	Pancreas	-0.00	-4.14	-3.09	-2.57	-2.07	-0.10	-0.03
ACH-000223	Breast	0.13	-0.20	-2.76	-2.20	-2.53	-0.08	0.22
ACH-000226	Lymphoid	0.19	-0.18	-1.60	-2.47	-2.11	-0.06	0.57
ACH-000227	Peripheral Nervous	0.07	-0.67	-2.97	-2.10	-2.60	-0.15	0.37
ACH-000228	Head and Neck	0.14	-0.35	-3.67	-1.79	-2.34	-0.12	0.03
ACH-000231	CNS/Brain	0.18	-0.29	-2.56	-2.57	-2.26	-0.20	0.15
ACH-000232	CNS/Brain	0.38	-0.27	-2.09	-1.86	-2.27	-0.20	0.34
ACH-000233	Lymphoid	0.00	-0.27	-2.67	-2.87	-2.51	-0.18	0.14

continued

Model	Lineage	CDKN2A	KRAS	PCNA	PSMA1	RPS3	SMAD4	TP53
ACH-000234	Kidney	0.25	-0.67	-2.39	-2.86	-2.72	0.04	0.63
ACH-000235	Pancreas	0.11	-3.00	-2.38	-2.44	-2.56	-0.14	0.21
ACH-000237	Ovary/Fallopian Tu	0.36	-0.30	-3.39	-2.33	-2.51	-0.27	0.18
ACH-000238	Head and Neck	-0.03	-0.29	-3.48	-2.03	-2.35	-0.17	0.05
ACH-000239	Esophagus/Stomach	-0.17	-0.27	-2.56	-2.32	-2.32	-0.35	0.23
ACH-000242	Bladder/Urinary Tr	0.01	-0.75	-2.58	-2.65	-2.57	0.11	0.14
ACH-000243	Pancreas	0.20	-1.67	-2.88	-2.20	-2.24	-0.17	-0.01
ACH-000244	CNS/Brain	0.15	-0.52	-1.27	-3.18	-1.86	0.04	2.30
ACH-000246	Kidney	0.10	-0.21	-1.68	-2.59	-3.08	-0.20	2.40
ACH-000247	Esophagus/Stomach	0.24	-0.20	-1.51	-1.53	-2.90	-0.57	0.70
ACH-000248	Breast	0.29	-0.32	-3.06	-2.66	-2.52	0.00	0.12

H GTEx v8 tissue medians for the panel (complete)

Median TPM by tissue for panel genes; the complete 432-row record. Background for the expression-specificity discussion in the body.

Table 14: GTEx median TPM by tissue, panel genes.

Gene	Tissue	Median TPM	<i>n</i>
KRAS	Thyroid	15.03	653
KRAS	Testis	10.87	361
KRAS	Small_Intestine_Terminal_Ileum	16.59	187
KRAS	Skin_Not_Sun_Exposed_Suprapubic	15.93	604
KRAS	Brain_Frontal_Cortex_BA9	16.95	209
KRAS	Fallopian_Tube	14.91	9
KRAS	Bladder	18.35	21
KRAS	Vagina	18.42	156
KRAS	Whole_Blood	9.09	755
KRAS	Brain_Amygdala	7.76	152
KRAS	Pituitary	9.70	283
KRAS	Minor_Salivary_Gland	14.30	162
KRAS	Adipose_Subcutaneous	14.68	663
KRAS	Adrenal_Gland	9.11	258
KRAS	Heart_Atrial_Appendage	5.41	429
KRAS	Breast_Mammary_Tissue	12.80	459
KRAS	Stomach	14.98	359
KRAS	Brain_Caudate_basal_ganglia	7.73	246
KRAS	Colon_Transverse	17.91	406
KRAS	Brain_Cerebellum	17.42	241
KRAS	Esophagus_Muscularis	22.17	515
KRAS	Liver	6.29	226
KRAS	Muscle_Skeletal	5.03	803
KRAS	Prostate	12.40	245
KRAS	Pancreas	5.03	328
KRAS	Nerve_Tibial	30.74	619
KRAS	Brain_Hippocampus	6.98	197
KRAS	Spleen	12.84	241
KRAS	Colon_Sigmoid	16.64	373
KRAS	Brain_Anterior_cingulate_cortex_BA24	11.62	176
KRAS	Esophagus_Gastroesophageal_Junction	20.19	375
KRAS	Brain_Hypothalamus	10.91	202
KRAS	Brain_Cortex	9.42	255
KRAS	Heart_Left_Ventricle	3.66	432

continued

Gene	Tissue	Median TPM	<i>n</i>
KRAS	Artery_Tibial	17.65	663
KRAS	Uterus	16.68	142
KRAS	Ovary	12.96	180
KRAS	Cells_EBV-transformed_lymphocytes	17.05	174
KRAS	Artery_Coronary	15.15	240
KRAS	Brain_Spinal_cord_cervical_c-1	7.86	159
KRAS	Adipose_Visceral_Omentum	12.77	541
KRAS	Brain_Nucleus_accumbens_basal_ganglia	9.27	246
KRAS	Brain_Cerebellar_Hemisphere	25.09	215
KRAS	Esophagus_Mucosa	21.55	555
KRAS	Cells_Cultured_fibroblasts	16.80	504
KRAS	Artery_Aorta	14.54	432
KRAS	Brain_Putamen_basal_ganglia	5.61	205
KRAS	Kidney_Medulla	9.44	4
KRAS	Lung	19.75	578
KRAS	Kidney_Cortex	6.95	85
KRAS	Brain_Substantia_nigra	6.82	139
KRAS	Cervix_Endocervix	13.31	10
KRAS	Skin_Sun_Exposed_Lower_leg	16.30	701
KRAS	Cervix_Ectocervix	16.20	9
TP53	Thyroid	19.66	653
TP53	Testis	14.83	361
TP53	Small_Intestine_Terminal_Ileum	18.42	187
TP53	Skin_Not_Sun_Exposed_Suprapubic	35.16	604
TP53	Brain_Frontal_Cortex_BA9	2.91	209
TP53	Fallopian_Tube	28.16	9
TP53	Bladder	23.77	21
TP53	Vagina	25.29	156
TP53	Whole_Blood	7.70	755
TP53	Brain_Amygdala	3.66	152
TP53	Pituitary	8.45	283
TP53	Minor_Salivary_Gland	18.84	162
TP53	Adipose_Subcutaneous	22.46	663
TP53	Adrenal_Gland	13.50	258
TP53	Heart_Atrial_Appendage	5.80	429
TP53	Breast_Mammary_Tissue	24.56	459
TP53	Stomach	12.63	359
TP53	Brain_Caudate_basal_ganglia	4.02	246
TP53	Colon_Transverse	17.20	406
TP53	Brain_Cerebellum	2.45	241
TP53	Esophagus_Muscularis	12.50	515
TP53	Liver	6.87	226
TP53	Muscle_Skeletal	5.44	803
TP53	Prostate	20.30	245
TP53	Pancreas	7.12	328
TP53	Nerve_Tibial	24.16	619
TP53	Brain_Hippocampus	2.71	197
TP53	Spleen	26.59	241
TP53	Colon_Sigmoid	14.54	373
TP53	Brain_Anterior_cingulate_cortex_BA24	3.13	176
TP53	Esophagus_Gastroesophageal_Junction	14.19	375
TP53	Brain_Hypothalamus	3.15	202
TP53	Brain_Cortex	3.65	255
TP53	Heart_Left_Ventricle	3.68	432
TP53	Artery_Tibial	20.53	663
TP53	Uterus	29.50	142
TP53	Ovary	32.43	180

continued

Gene	Tissue	Median TPM	<i>n</i>
TP53	Cells_EBV-transformed_lymphocytes	72.95	174
TP53	Artery_Coronary	19.73	240
TP53	Brain_Spinal_cord_cervical_c-1	4.16	159
TP53	Adipose_Visceral_Omentum	17.80	541
TP53	Brain_Nucleus_accumbens_basal_ganglia	3.37	246
TP53	Brain_Cerebellar_Hemisphere	2.03	215
TP53	Esophagus_Mucosa	25.84	555
TP53	Cells_Cultured_fibroblasts	32.87	504
TP53	Artery_Aorta	18.48	432
TP53	Brain_Putamen_basal_ganglia	3.32	205
TP53	Kidney_Medulla	13.02	4
TP53	Lung	16.92	578
TP53	Kidney_Cortex	5.96	85
TP53	Brain_Substantia_nigra	3.89	139
TP53	Cervix_Endocervix	29.50	10
TP53	Skin_Sun_Exposed_Lower_leg	36.96	701
TP53	Cervix_Ectocervix	28.48	9
CDKN2A	Thyroid	0.69	653
CDKN2A	Testis	7.13	361
CDKN2A	Small_Intestine_Terminal_Ileum	1.57	187
CDKN2A	Skin_Not_Sun_Exposed_Suprapubic	0.32	604
CDKN2A	Brain_Frontal_Cortex_BA9	0.62	209
CDKN2A	Fallopian_Tube	1.50	9
CDKN2A	Bladder	0.94	21
CDKN2A	Vagina	1.11	156
CDKN2A	Whole_Blood	0.28	755
CDKN2A	Brain_Amygdala	0.66	152
CDKN2A	Pituitary	16.91	283
CDKN2A	Minor_Salivary_Gland	0.60	162
CDKN2A	Adipose_Subcutaneous	0.53	663
CDKN2A	Adrenal_Gland	2.96	258
CDKN2A	Heart_Atrial_Appendage	0.07	429
CDKN2A	Breast_Mammary_Tissue	0.79	459
CDKN2A	Stomach	0.65	359
CDKN2A	Brain_Caudate_basal_ganglia	0.61	246
CDKN2A	Colon_Transverse	0.50	406
CDKN2A	Brain_Cerebellum	0.66	241
CDKN2A	Esophagus_Muscularis	1.17	515
CDKN2A	Liver	0.10	226
CDKN2A	Muscle_Skeletal	0.06	803
CDKN2A	Prostate	0.94	245
CDKN2A	Pancreas	0.22	328
CDKN2A	Nerve_Tibial	1.20	619
CDKN2A	Brain_Hippocampus	0.39	197
CDKN2A	Spleen	2.57	241
CDKN2A	Colon_Sigmoid	0.73	373
CDKN2A	Brain_Anterior_cingulate_cortex_BA24	0.38	176
CDKN2A	Esophagus_Gastroesophageal_Junction	0.70	375
CDKN2A	Brain_Hypothalamus	0.22	202
CDKN2A	Brain_Cortex	0.59	255
CDKN2A	Heart_Left_Ventricle	0.06	432
CDKN2A	Artery_Tibial	0.99	663
CDKN2A	Uterus	0.40	142
CDKN2A	Ovary	0.36	180
CDKN2A	Cells_EBV-transformed_lymphocytes	7.09	174
CDKN2A	Artery_Coronary	1.51	240
CDKN2A	Brain_Spinal_cord_cervical_c-1	0.78	159

continued

Gene	Tissue	Median TPM	<i>n</i>
CDKN2A	Adipose_Visceral_Omentum	0.58	541
CDKN2A	Brain_Nucleus_accumbens_basal_ganglia	0.27	246
CDKN2A	Brain_Cerebellar_Hemisphere	0.32	215
CDKN2A	Esophagus_Mucosa	2.12	555
CDKN2A	Cells_Cultured_fibroblasts	8.06	504
CDKN2A	Artery_Aorta	1.53	432
CDKN2A	Brain_Putamen_basal_ganglia	0.70	205
CDKN2A	Kidney_Medulla	0.15	4
CDKN2A	Lung	1.08	578
CDKN2A	Kidney_Cortex	0.09	85
CDKN2A	Brain_Substantia_nigra	0.32	139
CDKN2A	Cervix_Endocervix	0.73	10
CDKN2A	Skin_Sun_Exposed_Lower_leg	0.36	701
CDKN2A	Cervix_Ectocervix	0.61	9
SMAD4	Thyroid	31.22	653
SMAD4	Testis	17.37	361
SMAD4	Small_Intestine_Terminal_Ileum	19.95	187
SMAD4	Skin_Not_Sun_Exposed_Suprapubic	19.61	604
SMAD4	Brain_Frontal_Cortex_BA9	7.55	209
SMAD4	Fallopian_Tube	32.06	9
SMAD4	Bladder	26.01	21
SMAD4	Vagina	23.68	156
SMAD4	Whole_Blood	5.09	755
SMAD4	Brain_Amygdala	6.10	152
SMAD4	Pituitary	21.59	283
SMAD4	Minor_Salivary_Gland	20.42	162
SMAD4	Adipose_Subcutaneous	25.15	663
SMAD4	Adrenal_Gland	14.26	258
SMAD4	Heart_Atrial_Appendage	10.39	429
SMAD4	Breast_Mammary_Tissue	23.28	459
SMAD4	Stomach	15.23	359
SMAD4	Brain_Caudate_basal_ganglia	7.37	246
SMAD4	Colon_Transverse	18.37	406
SMAD4	Brain_Cerebellum	28.79	241
SMAD4	Esophagus_Muscularis	27.33	515
SMAD4	Liver	9.22	226
SMAD4	Muscle_Skeletal	18.43	803
SMAD4	Prostate	26.64	245
SMAD4	Pancreas	8.75	328
SMAD4	Nerve_Tibial	32.34	619
SMAD4	Brain_Hippocampus	6.37	197
SMAD4	Spleen	18.52	241
SMAD4	Colon_Sigmoid	28.02	373
SMAD4	Brain_Anterior_cingulate_cortex_BA24	6.54	176
SMAD4	Esophagus_Gastroesophageal_Junction	26.77	375
SMAD4	Brain_Hypothalamus	7.31	202
SMAD4	Brain_Cortex	7.25	255
SMAD4	Heart_Left_Ventricle	7.63	432
SMAD4	Artery_Tibial	30.44	663
SMAD4	Uterus	37.22	142
SMAD4	Ovary	34.94	180
SMAD4	Cells_EBV-transformed_lymphocytes	16.66	174
SMAD4	Artery_Coronary	24.00	240
SMAD4	Brain_Spinal_cord_cervical_c-1	8.84	159
SMAD4	Adipose_Visceral_Omentum	19.44	541
SMAD4	Brain_Nucleus_accumbens_basal_ganglia	8.14	246
SMAD4	Brain_Cerebellar_Hemisphere	32.46	215

continued

Gene	Tissue	Median TPM	<i>n</i>
SMAD4	Esophagus_Mucosa	12.92	555
SMAD4	Cells_Cultured_fibroblasts	23.69	504
SMAD4	Artery_Aorta	24.20	432
SMAD4	Brain_Putamen_basal_ganglia	5.95	205
SMAD4	Kidney_Medulla	13.07	4
SMAD4	Lung	24.98	578
SMAD4	Kidney_Cortex	9.28	85
SMAD4	Brain_Substantia_nigra	6.20	139
SMAD4	Cervix_Endocervix	34.52	10
SMAD4	Skin_Sun_Exposed_Lower_leg	19.66	701
SMAD4	Cervix_Ectocervix	32.24	9
GAPDH	Thyroid	812.90	653
GAPDH	Testis	593.40	361
GAPDH	Small_Intestine_Terminal_Ileum	866.60	187
GAPDH	Skin_Not_Sun_Exposed_Suprapubic	928.55	604
GAPDH	Brain_Frontal_Cortex_BA9	1651.00	209
GAPDH	Fallopian_Tube	956.30	9
GAPDH	Bladder	1394.00	21
GAPDH	Vagina	1209.00	156
GAPDH	Whole_Blood	1774.00	755
GAPDH	Brain_Amygdala	1002.50	152
GAPDH	Pituitary	919.30	283
GAPDH	Minor_Salivary_Gland	945.00	162
GAPDH	Adipose_Subcutaneous	831.80	663
GAPDH	Adrenal_Gland	905.40	258
GAPDH	Heart_Atrial_Appendage	1533.00	429
GAPDH	Breast_Mammary_Tissue	721.50	459
GAPDH	Stomach	528.20	359
GAPDH	Brain_Caudate_basal_ganglia	1074.00	246
GAPDH	Colon_Transverse	947.50	406
GAPDH	Brain_Cerebellum	1198.00	241
GAPDH	Esophagus_Muscularis	1393.00	515
GAPDH	Liver	512.05	226
GAPDH	Muscle_Skeletal	5597.00	803
GAPDH	Prostate	787.00	245
GAPDH	Pancreas	147.60	328
GAPDH	Nerve_Tibial	1126.00	619
GAPDH	Brain_Hippocampus	893.60	197
GAPDH	Spleen	716.40	241
GAPDH	Colon_Sigmoid	1094.00	373
GAPDH	Brain_Anterior_cingulate_cortex_BA24	1183.50	176
GAPDH	Esophagus_Gastroesophageal_Junction	1301.00	375
GAPDH	Brain_Hypothalamus	1198.50	202
GAPDH	Brain_Cortex	1439.00	255
GAPDH	Heart_Left_Ventricle	1490.00	432
GAPDH	Artery_Tibial	1523.00	663
GAPDH	Uterus	904.35	142
GAPDH	Ovary	831.30	180
GAPDH	Cells_EBV-transformed_lymphocytes	6385.00	174
GAPDH	Artery_Coronary	1574.00	240
GAPDH	Brain_Spinal_cord_cervical_c-1	1403.00	159
GAPDH	Adipose_Visceral_Omentum	705.20	541
GAPDH	Brain_Nucleus_accumbens_basal_ganglia	1141.50	246
GAPDH	Brain_Cerebellar_Hemisphere	1259.00	215
GAPDH	Esophagus_Mucosa	1657.00	555
GAPDH	Cells_Cultured_fibroblasts	4452.50	504
GAPDH	Artery_Aorta	2012.00	432

continued

Gene	Tissue	Median TPM	<i>n</i>
GAPDH	Brain_Putamen_basal_ganglia	908.50	205
GAPDH	Kidney_Medulla	765.55	4
GAPDH	Lung	852.85	578
GAPDH	Kidney_Cortex	803.20	85
GAPDH	Brain_Substantia_nigra	1084.00	139
GAPDH	Cervix_Endocervix	927.35	10
GAPDH	Skin_Sun_Exposed_Lower_leg	1119.00	701
GAPDH	Cervix_Ectocervix	727.60	9
ACTB	Thyroid	2256.00	653
ACTB	Testis	1018.00	361
ACTB	Small_Intestine_Terminal_Ileum	3378.00	187
ACTB	Skin_Not_Sun_Exposed_Suprapubic	1995.50	604
ACTB	Brain_Frontal_Cortex_BA9	1525.00	209
ACTB	Fallopian_Tube	5738.00	9
ACTB	Bladder	6552.00	21
ACTB	Vagina	3645.00	156
ACTB	Whole_Blood	4804.00	755
ACTB	Brain_Amygdala	1011.50	152
ACTB	Pituitary	1204.00	283
ACTB	Minor_Salivary_Gland	1726.50	162
ACTB	Adipose_Subcutaneous	3523.00	663
ACTB	Adrenal_Gland	1439.50	258
ACTB	Heart_Atrial_Appendage	1200.00	429
ACTB	Breast_Mammary_Tissue	3130.00	459
ACTB	Stomach	1996.00	359
ACTB	Brain_Caudate_basal_ganglia	1181.00	246
ACTB	Colon_Transverse	4372.50	406
ACTB	Brain_Cerebellum	1205.00	241
ACTB	Esophagus_Muscularis	9782.00	515
ACTB	Liver	760.45	226
ACTB	Muscle_Skeletal	396.00	803
ACTB	Prostate	3107.00	245
ACTB	Pancreas	404.80	328
ACTB	Nerve_Tibial	3386.00	619
ACTB	Brain_Hippocampus	1107.00	197
ACTB	Spleen	3838.00	241
ACTB	Colon_Sigmoid	10180.00	373
ACTB	Brain_Anterior_cingulate_cortex_BA24	1186.00	176
ACTB	Esophagus_Gastroesophageal_Junction	8623.00	375
ACTB	Brain_Hypothalamus	1026.50	202
ACTB	Brain_Cortex	1310.00	255
ACTB	Heart_Left_Ventricle	452.00	432
ACTB	Artery_Tibial	11600.00	663
ACTB	Uterus	7323.00	142
ACTB	Ovary	2810.50	180
ACTB	Cells_EBV-transformed_lymphocytes	11630.00	174
ACTB	Artery_Coronary	8362.00	240
ACTB	Brain_Spinal_cord_cervical_c-1	2144.00	159
ACTB	Adipose_Visceral_Omentum	3369.00	541
ACTB	Brain_Nucleus_accumbens_basal_ganglia	1228.00	246
ACTB	Brain_Cerebellar_Hemisphere	1240.00	215
ACTB	Esophagus_Mucosa	3467.00	555
ACTB	Cells_Cultured_fibroblasts	6699.00	504
ACTB	Artery_Aorta	9154.00	432
ACTB	Brain_Putamen_basal_ganglia	970.70	205
ACTB	Kidney_Medulla	1490.50	4
ACTB	Lung	4809.50	578

continued

Gene	Tissue	Median TPM	<i>n</i>
ACTB	Kidney_Cortex	1066.00	85
ACTB	Brain_Substantia_nigra	1020.00	139
ACTB	Cervix_Endocervix	4252.00	10
ACTB	Skin_Sun_Exposed_Lower_leg	2169.00	701
ACTB	Cervix_Ectocervix	3832.00	9
PRSS1	Thyroid	2.56	653
PRSS1	Testis	6.26	361
PRSS1	Small_Intestine_Terminal_Ileum	7.43	187
PRSS1	Skin_Not_Sun_Exposed_Suprapubic	1.55	604
PRSS1	Brain_Frontal_Cortex_BA9	0.60	209
PRSS1	Fallopian_Tube	13.89	9
PRSS1	Bladder	8.71	21
PRSS1	Vagina	6.08	156
PRSS1	Whole_Blood	0.24	755
PRSS1	Brain_Amygdala	0.09	152
PRSS1	Pituitary	7.24	283
PRSS1	Minor_Salivary_Gland	3.03	162
PRSS1	Adipose_Subcutaneous	1.27	663
PRSS1	Adrenal_Gland	7.38	258
PRSS1	Heart_Atrial_Appendage	0.97	429
PRSS1	Breast_Mammary_Tissue	1.98	459
PRSS1	Stomach	28.36	359
PRSS1	Brain_Caudate_basal_ganglia	0.17	246
PRSS1	Colon_Transverse	4.04	406
PRSS1	Brain_Cerebellum	5.70	241
PRSS1	Esophagus_Muscularis	4.24	515
PRSS1	Liver	7.05	226
PRSS1	Muscle_Skeletal	0.38	803
PRSS1	Prostate	6.26	245
PRSS1	Pancreas	78675.00	328
PRSS1	Nerve_Tibial	1.62	619
PRSS1	Brain_Hippocampus	0.39	197
PRSS1	Spleen	10.22	241
PRSS1	Colon_Sigmoid	1.65	373
PRSS1	Brain_Anterior_cingulate_cortex_BA24	0.28	176
PRSS1	Esophagus_Gastroesophageal_Junction	1.71	375
PRSS1	Brain_Hypothalamus	0.38	202
PRSS1	Brain_Cortex	0.39	255
PRSS1	Heart_Left_Ventricle	0.82	432
PRSS1	Artery_Tibial	0.71	663
PRSS1	Uterus	10.03	142
PRSS1	Ovary	11.42	180
PRSS1	Cells_EBV-transformed_lymphocytes	5.25	174
PRSS1	Artery_Coronary	3.83	240
PRSS1	Brain_Spinal_cord_cervical_c-1	0.51	159
PRSS1	Adipose_Visceral_Omentum	2.53	541
PRSS1	Brain_Nucleus_accumbens_basal_ganglia	0.08	246
PRSS1	Brain_Cerebellar_Hemisphere	1.15	215
PRSS1	Esophagus_Mucosa	7.68	555
PRSS1	Cells_Cultured_fibroblasts	1.18	504
PRSS1	Artery_Aorta	4.56	432
PRSS1	Brain_Putamen_basal_ganglia	0.09	205
PRSS1	Kidney_Medulla	4.50	4
PRSS1	Lung	2.59	578
PRSS1	Kidney_Cortex	3.64	85
PRSS1	Brain_Substantia_nigra	0.29	139
PRSS1	Cervix_Endocervix	6.95	10

continued

Gene	Tissue	Median TPM	n
PRSS1	Skin_Sun_Exposed_Lower_leg	1.07	701
PRSS1	Cervix_Ectocervix	14.16	9
INS	Thyroid	0.11	653
INS	Testis	0.27	361
INS	Small_Intestine_Terminal_Ileum	0.19	187
INS	Skin_Not_Sun_Exposed_Suprapubic	0.04	604
INS	Brain_Frontal_Cortex_BA9	0	209
INS	Fallopian_Tube	0.31	9
INS	Bladder	0.18	21
INS	Vagina	0.17	156
INS	Whole_Blood	0	755
INS	Brain_Amygdala	0	152
INS	Pituitary	0.25	283
INS	Minor_Salivary_Gland	0.11	162
INS	Adipose_Subcutaneous	0	663
INS	Adrenal_Gland	0.47	258
INS	Heart_Atrial_Appendage	0	429
INS	Breast_Mammary_Tissue	0	459
INS	Stomach	0.19	359
INS	Brain_Caudate_basal_ganglia	0.04	246
INS	Colon_Transverse	0.12	406
INS	Brain_Cerebellum	0.16	241
INS	Esophagus_Muscularis	0.10	515
INS	Liver	0.26	226
INS	Muscle_Skeletal	0.02	803
INS	Prostate	0.17	245
INS	Pancreas	2325.50	328
INS	Nerve_Tibial	0.08	619
INS	Brain_Hippocampus	0	197
INS	Spleen	0.34	241
INS	Colon_Sigmoid	0.07	373
INS	Brain_Anterior_cingulate_cortex_BA24	0	176
INS	Esophagus_Gastroesophageal_Junction	0.04	375
INS	Brain_Hypothalamus	0.25	202
INS	Brain_Cortex	0.03	255
INS	Heart_Left_Ventricle	0.02	432
INS	Artery_Tibial	0	663
INS	Uterus	0.26	142
INS	Ovary	0.31	180
INS	Cells_EBV-transformed_lymphocytes	0.14	174
INS	Artery_Coronary	0.12	240
INS	Brain_Spinal_cord_cervical_c-1	0	159
INS	Adipose_Visceral_Omentum	0.09	541
INS	Brain_Nucleus_accumbens_basal_ganglia	0	246
INS	Brain_Cerebellar_Hemisphere	0.09	215
INS	Esophagus_Mucosa	0.21	555
INS	Cells_Cultured_fibroblasts	0.04	504
INS	Artery_Aorta	0.13	432
INS	Brain_Putamen_basal_ganglia	0	205
INS	Kidney_Medulla	0.10	4
INS	Lung	0.08	578
INS	Kidney_Cortex	0.13	85
INS	Brain_Substantia_nigra	0.44	139
INS	Cervix_Endocervix	0.16	10
INS	Skin_Sun_Exposed_Lower_leg	0.06	701
INS	Cervix_Ectocervix	0.45	9